Retail Sales and Marketing Campaign Data Analysis using Excel, Power BI, MySQL , and Python

AXIA COHORT 8 PROJ ECT

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Executive Summary

This project was developed as part of the Axia Cohort 8 Data Analytics Training to demonstrate practical skills in Excel, Power BI, MySQL, and Python for solving real-world business problems using retail sales and marketing data.

Introduction

The analysis followed four stages:

- Excel: Used for initial data cleaning, descriptive analysis, and validation.
- Power BI: Designed an interactive dashboard to visualize key metrics such as ROI, Ad Spend, Conversion Rate, and Revenue trends.
- MySQL: Built and queried a structured retail database to answer critical business questions, such as top-selling categories, high-value customers, and seasonal sales performance.
- Python (Pandas & Matplotlib): Applied advanced data analysis, statistical measures, and visualizations to uncover deeper insights into revenue trends, customer behavior, and product performance.

Excel Data Cleaning Report

Project Overview

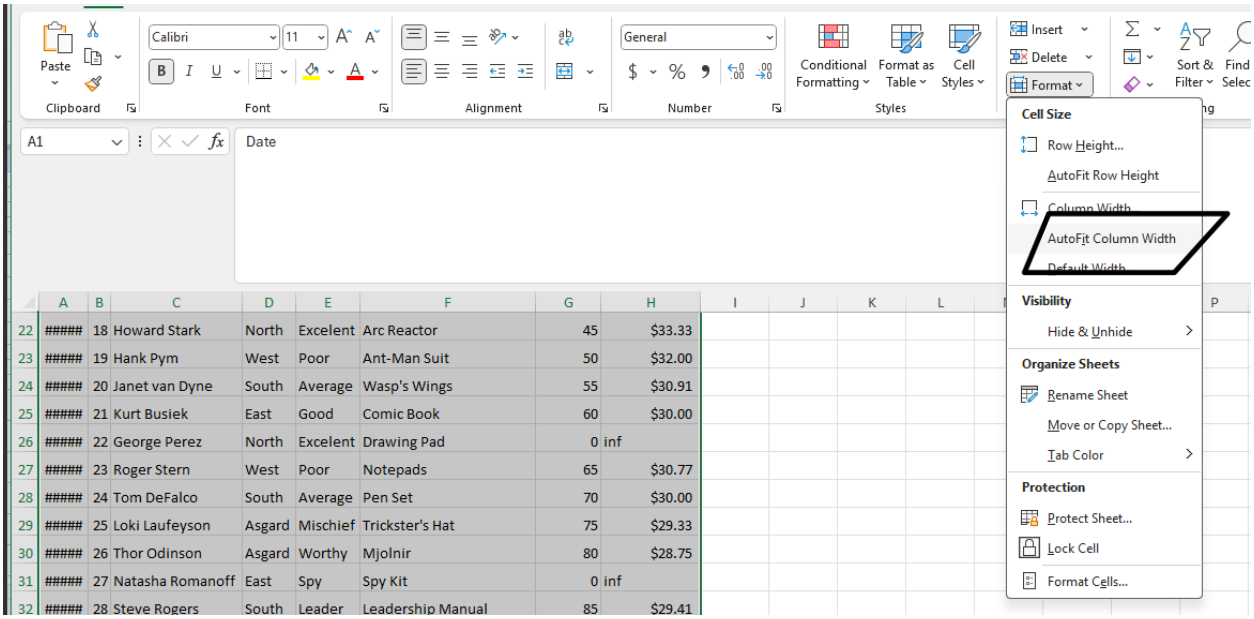
This report outlines the data cleaning procedures carried out on a messy dataset using **Microsoft Excel**. The dataset simulated common real-world data issues such as inconsistent entries, extra spaces, missing values, duplicates, and formatting errors. The following cleaning techniques were applied to ensure the data is accurate, consistent, and analysis-ready.

1. Autofit Columns and Rows

To enhance readability and avoid hidden data:

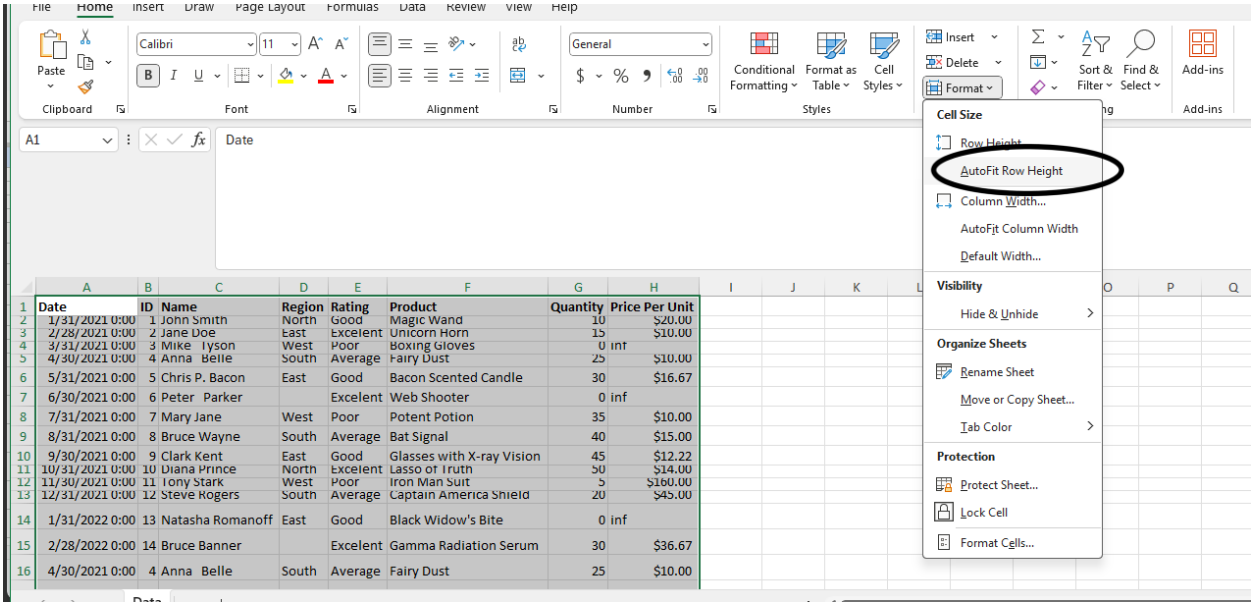
- All columns and rows were **autofitted** by clicking on Ctrl+A to select the entire sheet, then the Home tab, click on format (Found in the cells group on the right), choose Autofit Column width, this adjusted the column widths to fit the content. Then click format again and choose Autofit Row Height.
- This ensured that data entries were fully visible without manual resizing.

AUTOFIT COLUMN WIDTH



	A	B	C	D	E	F	G	H
22	#####	18	Howard Stark	North	Excelent	Arc Reactor	45	$33.33
23	#####	19	Hank Pym	West	Poor	Ant-Man Suit	50	$32.00
24	#####	20	Janet van Dyne	South	Average	Wasp's Wings	55	$30.91
25	#####	21	Kurt Busiek	East	Good	Comic Book	60	$30.00
26	#####	22	George Perez	North	Excelent	Drawing Pad	0	inf
27	#####	23	Roger Stern	West	Poor	Notepads	65	$30.77
28	#####	24	Tom DeFalco	South	Average	Pen Set	70	$30.00
29	#####	25	Loki Laufeyson	Asgard	Mischief	Trickster's Hat	75	$29.33
30	#####	26	Thor Odinson	Asgard	Worthy	Mjolnir	80	$28.75
31	#####	27	Natasha Romanoff	East	Spy	Spy Kit	0	inf
32	#####	28	Steve Rogers	South	Leader	Leadership Manual	85	$29.41

AUTOFIT ROW HEIGHT

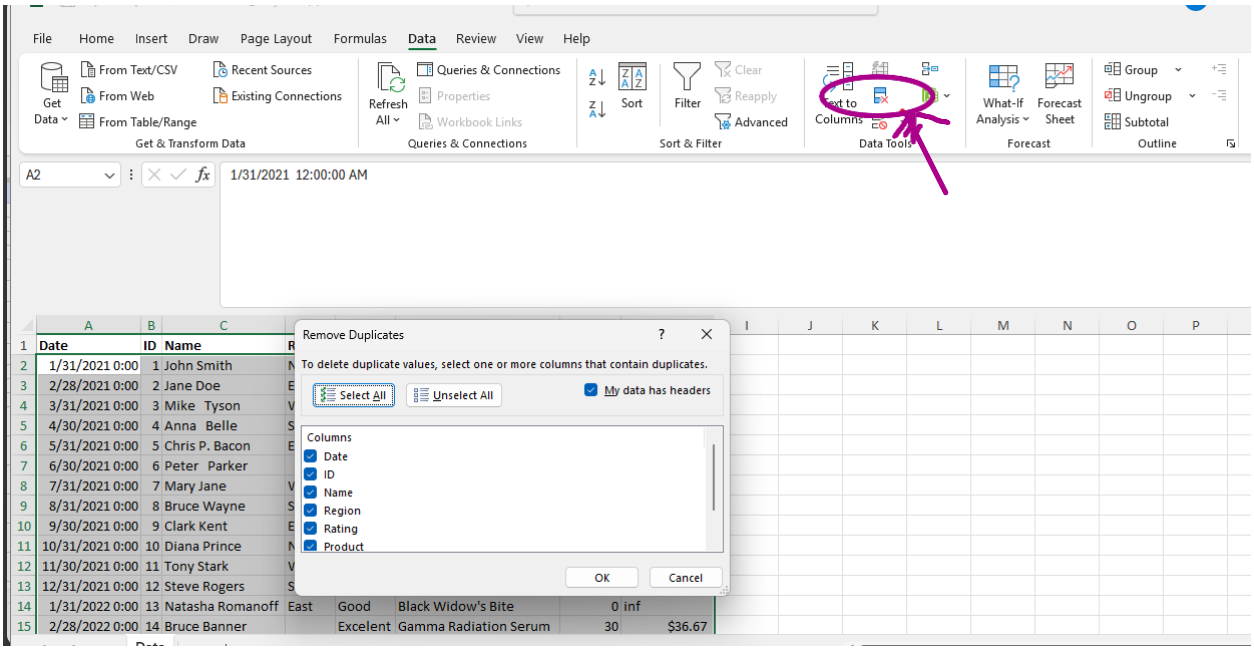


Date	ID	Name	Region	Rating	Product	Quantity	Price Per Unit
1/31/2021 0:00	1	John Smith	North	Good	Magic Wand	10	$20.00
2/28/2021 0:00	2	Jane Doe	East	Excelent	Unicorn Horn	15	$10.00
3/31/2021 0:00	3	Mike Tyson	West	Poor	Boxing Gloves	0	inf
4/30/2021 0:00	4	Anna Belle	South	Average	Fairy Dust	25	$10.00
5/31/2021 0:00	5	Chris P. Bacon	East	Good	Bacon Scented Candle	30	$16.67
6/30/2021 0:00	6	Peter Parker		Excelent	Web Shooter	0	inf
7/31/2021 0:00	7	Mary Jane	West	Poor	Potent Potion	35	$10.00
8/31/2021 0:00	8	Bruce Wayne	South	Average	Bat Signal	40	$15.00
9/30/2021 0:00	9	Clark Kent	East	Good	Glasses with X-ray Vision	45	$12.22
10/31/2021 0:00	10	Diana Prince	North	Excelent	Lasso of Truth	50	$14.00
11/30/2021 0:00	11	Tony Stark	West	Poor	Iron Man Suit	5	$160.00
12/31/2021 0:00	12	Steve Rogers	South	Average	Captain America Shield	20	$45.00
1/31/2022 0:00	13	Natasha Romanoff	East	Good	Black Widow's Bite	0	inf
2/28/2022 0:00	14	Bruce Banner		Excelent	Gamma Radiation Serum	30	$36.67
4/30/2021 0:00	4	Anna Belle	South	Average	Fairy Dust	25	$10.00

2. Identify and Remove Duplicates

To eliminate redundant records:

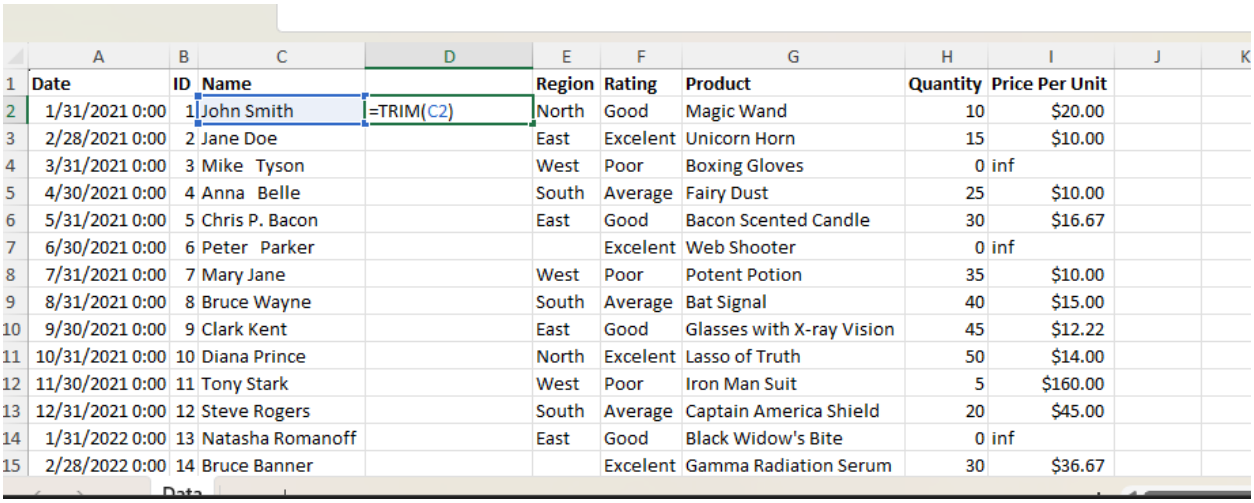
- Excel's **Remove Duplicates** feature (under the Data tab) was used. By selecting all the data, clicked the Data tab, clicked on remove duplicates, a window pops up showing the column names, made sure all columns were selected, clicked OK, All duplicates were removed.
- A full row match was checked across all relevant columns.
- Duplicate rows were deleted, leaving only unique data.



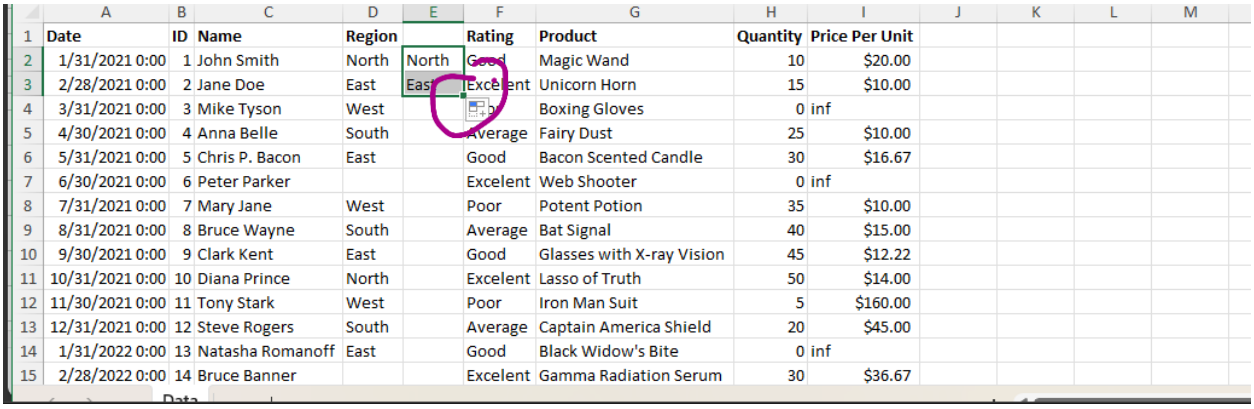
3. Trim Extra Spaces

To correct inconsistent formatting:

- The TRIM() function was applied to all text-based columns (Name, Region, Rating and Product Columns): Inserted a new column next to each text-based columns, in the first cell of the new column, typed =TRIM(C2), Clicked Enter then clicked on the small square at the bottom-right of the cell and dragged it down to apply the formula to all rows, once done, copied the entire new column, right-clicked the original Name column paste Special, then Values and then deleted the helper column used to TRIM. Repeated this process for other text-based columns like Region, Rating and Product.
- This removed **leading, trailing, and excess internal spaces**, which often cause hidden data mismatches.



	A	B	C	D	E	F	G	H	I	J	K
1	Date	ID	Name		Region	Rating	Product	Quantity	Price Per Unit		
2	1/31/2021 0:00	1	John Smith	=TRIM(C2)	North	Good	Magic Wand	10	$20.00		
3	2/28/2021 0:00	2	Jane Doe		East	Excelent	Unicorn Horn	15	$10.00		
4	3/31/2021 0:00	3	Mike Tyson		West	Poor	Boxing Gloves	0	inf		
5	4/30/2021 0:00	4	Anna Belle		South	Average	Fairy Dust	25	$10.00		
6	5/31/2021 0:00	5	Chris P. Bacon		East	Good	Bacon Scented Candle	30	$16.67		
7	6/30/2021 0:00	6	Peter Parker			Excelent	Web Shooter	0	inf		
8	7/31/2021 0:00	7	Mary Jane		West	Poor	Potent Potion	35	$10.00		
9	8/31/2021 0:00	8	Bruce Wayne		South	Average	Bat Signal	40	$15.00		
10	9/30/2021 0:00	9	Clark Kent		East	Good	Glasses with X-ray Vision	45	$12.22		
11	10/31/2021 0:00	10	Diana Prince		North	Excelent	Lasso of Truth	50	$14.00		
12	11/30/2021 0:00	11	Tony Stark		West	Poor	Iron Man Suit	5	$160.00		
13	12/31/2021 0:00	12	Steve Rogers		South	Average	Captain America Shield	20	$45.00		
14	1/31/2022 0:00	13	Natasha Romanoff		East	Good	Black Widow's Bite	0	inf		
15	2/28/2022 0:00	14	Bruce Banner			Excelent	Gamma Radiation Serum	30	$36.67		



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Date	ID	Name	Region		Rating	Product	Quantity	Price Per Unit				
2	1/31/2021 0:00	1	John Smith	North	North	Good	Magic Wand	10	$20.00				
3	2/28/2021 0:00	2	Jane Doe	East	East	Excellent	Unicorn Horn	15	$10.00				
4	3/31/2021 0:00	3	Mike Tyson	West			Boxing Gloves	0	inf				
5	4/30/2021 0:00	4	Anna Belle	South		Average	Fairy Dust	25	$10.00				
6	5/31/2021 0:00	5	Chris P. Bacon	East		Good	Bacon Scented Candle	30	$16.67				
7	6/30/2021 0:00	6	Peter Parker			Excelent	Web Shooter	0	inf				
8	7/31/2021 0:00	7	Mary Jane	West		Poor	Potent Potion	35	$10.00				
9	8/31/2021 0:00	8	Bruce Wayne	South		Average	Bat Signal	40	$15.00				
10	9/30/2021 0:00	9	Clark Kent	East		Good	Glasses with X-ray Vision	45	$12.22				
11	10/31/2021 0:00	10	Diana Prince	North		Excelent	Lasso of Truth	50	$14.00				
12	11/30/2021 0:00	11	Tony Stark	West		Poor	Iron Man Suit	5	$160.00				
13	12/31/2021 0:00	12	Steve Rogers	South		Average	Captain America Shield	20	$45.00				
14	1/31/2022 0:00	13	Natasha Romanoff	East		Good	Black Widow's Bite	0	inf				
15	2/28/2022 0:00	14	Bruce Banner			Excelent	Gamma Radiation Serum	30	$36.67				

4. Eliminate Blank Cells

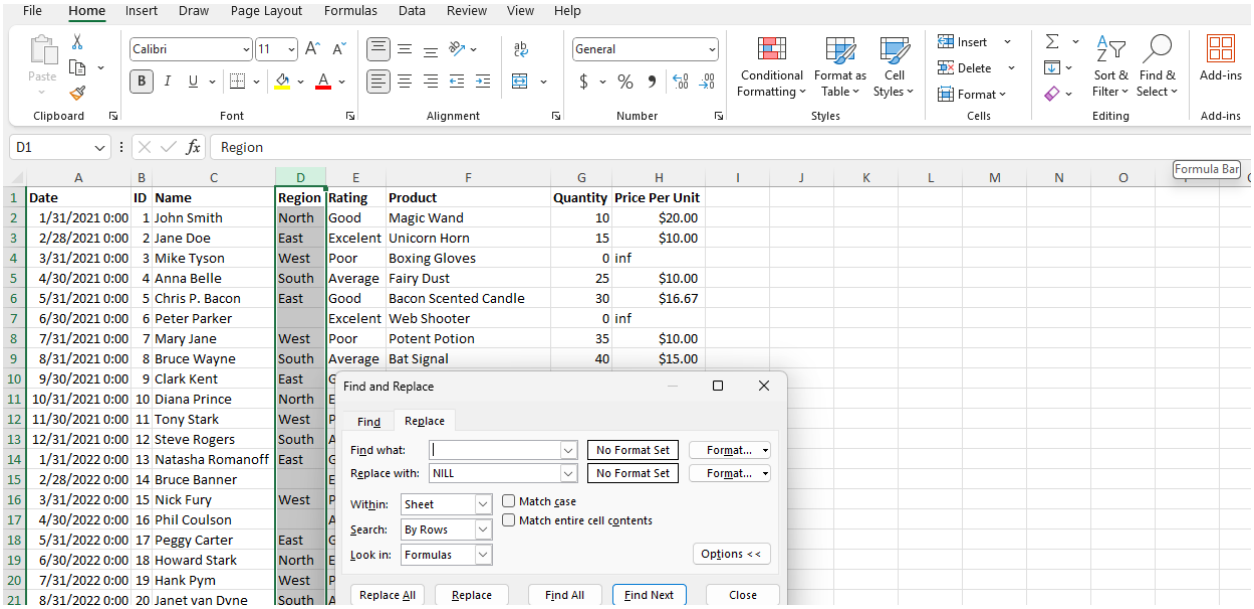
To ensure completeness:

- Filters were applied to locate blank rows and cells.
- Empty rows were deleted.
- In cases where data was expected but missing, placeholders like **"Nill"** or **"0"** were used to maintain structure.

5. Use Find and Replace to Correct Errors

To fix common text inconsistencies:

- The **Find and Replace tool (Ctrl + H)** was used to:
    - Fix missing words Or Figures: (In the find what box, inf was typed, and in the Replace with box $0 was typed, clicked Replace All, a window appears on Excel telling me how many changes were made, then clicked OK.
    - This was done for all missing words or figures.
    - Correct date formats was used

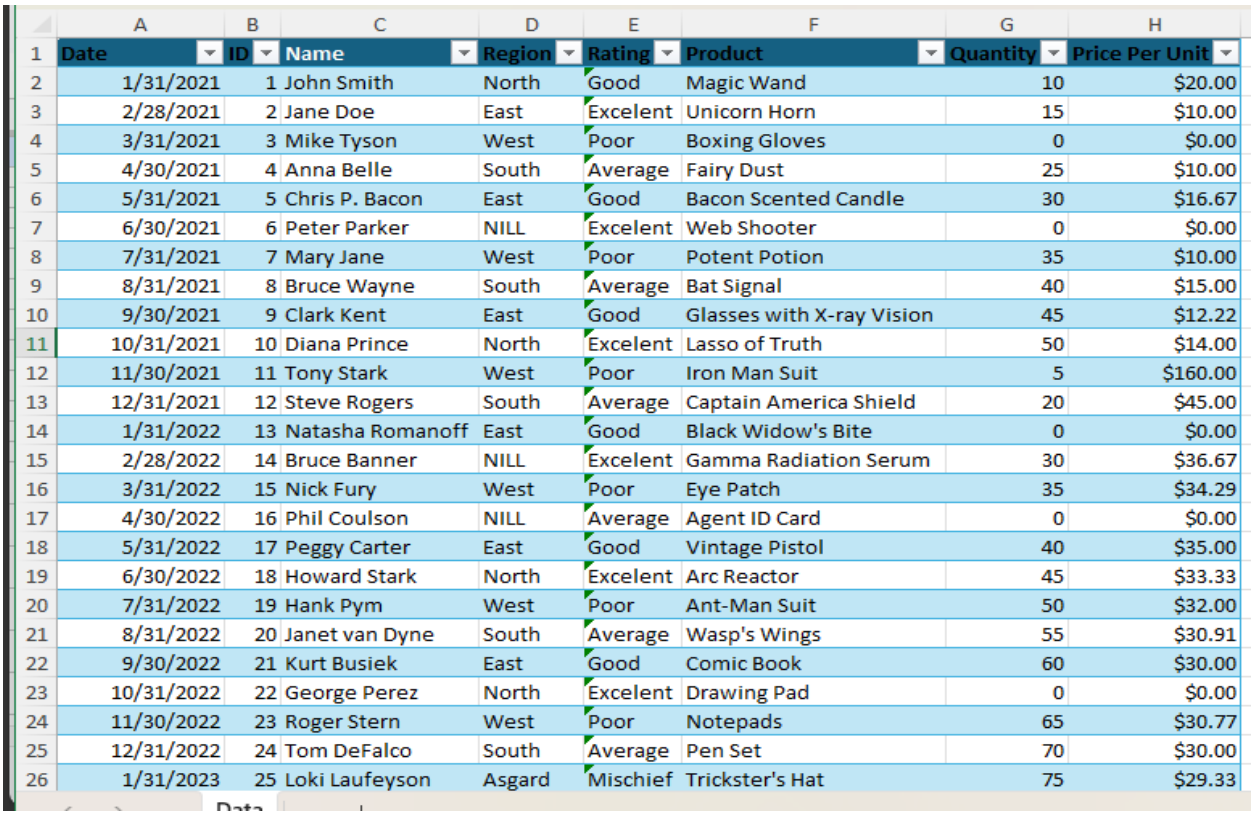


	A	B	C	D	E	F	G	H
1	Date	ID	Name	Region	Rating	Product	Quantity	Price Per Unit
2	1/31/2021 0:00	1	John Smith	North	Good	Magic Wand	10	$20.00
3	2/28/2021 0:00	2	Jane Doe	East	Excelent	Unicorn Horn	15	$10.00
4	3/31/2021 0:00	3	Mike Tyson	West	Poor	Boxing Gloves	0	inf
5	4/30/2021 0:00	4	Anna Belle	South	Average	Fairy Dust	25	$10.00
6	5/31/2021 0:00	5	Chris P. Bacon	East	Good	Bacon Scented Candle	30	$16.67
7	6/30/2021 0:00	6	Peter Parker		Excelent	Web Shooter	0	inf
8	7/31/2021 0:00	7	Mary Jane	West	Poor	Potent Potion	35	$10.00
9	8/31/2021 0:00	8	Bruce Wayne	South	Average	Bat Signal	40	$15.00
10	9/30/2021 0:00	9	Clark Kent	East				
11	10/31/2021 0:00	10	Diana Prince	North				
12	11/30/2021 0:00	11	Tony Stark	West				
13	12/31/2021 0:00	12	Steve Rogers	South				
14	1/31/2022 0:00	13	Natasha Romanoff	East				
15	2/28/2022 0:00	14	Bruce Banner					
16	3/31/2022 0:00	15	Nick Fury	West				
17	4/30/2022 0:00	16	Phil Coulson					
18	5/31/2022 0:00	17	Peggy Carter	East				
19	6/30/2022 0:00	18	Howard Stark	North				
20	7/31/2022 0:00	19	Hank Pym	West				
21	8/31/2022 0:00	20	Janet van Dyne	South				

6. Convert Data into Table

To enable structured data manipulation:

- The cleaned dataset was converted into a **formal Excel Table** via **Insert tab, then Table**.
- Tables allow dynamic filtering, sorting, and easy application of formulas.
- Column headers were named clearly and consistently.



Date	ID	Name	Region	Rating	Product	Quantity	Price Per Unit
1/31/2021	1	John Smith	North	Good	Magic Wand	10	$20.00
2/28/2021	2	Jane Doe	East	Excelent	Unicorn Horn	15	$10.00
3/31/2021	3	Mike Tyson	West	Poor	Boxing Gloves	0	$0.00
4/30/2021	4	Anna Belle	South	Average	Fairy Dust	25	$10.00
5/31/2021	5	Chris P. Bacon	East	Good	Bacon Scented Candle	30	$16.67
6/30/2021	6	Peter Parker	NILL	Excelent	Web Shooter	0	$0.00
7/31/2021	7	Mary Jane	West	Poor	Potent Potion	35	$10.00
8/31/2021	8	Bruce Wayne	South	Average	Bat Signal	40	$15.00
9/30/2021	9	Clark Kent	East	Good	Glasses with X-ray Vision	45	$12.22
10/31/2021	10	Diana Prince	North	Excelent	Lasso of Truth	50	$14.00
11/30/2021	11	Tony Stark	West	Poor	Iron Man Suit	5	$160.00
12/31/2021	12	Steve Rogers	South	Average	Captain America Shield	20	$45.00
1/31/2022	13	Natasha Romanoff	East	Good	Black Widow's Bite	0	$0.00
2/28/2022	14	Bruce Banner	NILL	Excelent	Gamma Radiation Serum	30	$36.67
3/31/2022	15	Nick Fury	West	Poor	Eye Patch	35	$34.29
4/30/2022	16	Phil Coulson	NILL	Average	Agent ID Card	0	$0.00
5/31/2022	17	Peggy Carter	East	Good	Vintage Pistol	40	$35.00
6/30/2022	18	Howard Stark	North	Excelent	Arc Reactor	45	$33.33
7/31/2022	19	Hank Pym	West	Poor	Ant-Man Suit	50	$32.00
8/31/2022	20	Janet van Dyne	South	Average	Wasp's Wings	55	$30.91
9/30/2022	21	Kurt Busiek	East	Good	Comic Book	60	$30.00
10/31/2022	22	George Perez	North	Excelent	Drawing Pad	0	$0.00
11/30/2022	23	Roger Stern	West	Poor	Notepads	65	$30.77
12/31/2022	24	Tom DeFalco	South	Average	Pen Set	70	$30.00
1/31/2023	25	Loki Laufeyson	Asgard	Mischief	Trickster's Hat	75	$29.33

7. Validate Data Thoroughly

To ensure consistency and integrity:

- **Data Validation** was applied:
    - Dropdown lists were created for categorical fields (e.g., Rating).
    - Date fields were restricted to valid date ranges.
    - Numeric fields were checked to prevent text or invalid entries.
- Inconsistencies were flagged using conditional formatting and logical checks.

Conclusion

The dataset has now been fully cleaned and structured. All major inconsistencies, blanks, and formatting issues have been resolved. It is now ready for accurate and efficient analysis in Excel or other analytical tools.

This exercise demonstrates the critical importance of data cleaning in ensuring high-quality insights and avoiding errors in downstream reporting or decision-making.

Power BI Marketing Campaign Dashboard Report

Project Overview

This Power BI project analyzed marketing campaign performance data to evaluate the effectiveness of different channels, categories, and products. Using DAX measures, visualizations, and slicers, key performance indicators such as ROI, Conversion Rate, Revenue, and Ad Spend were examined. The dashboard provides an interactive way for management to track marketing performance and support decision-making.

Step-by-Step Development Process

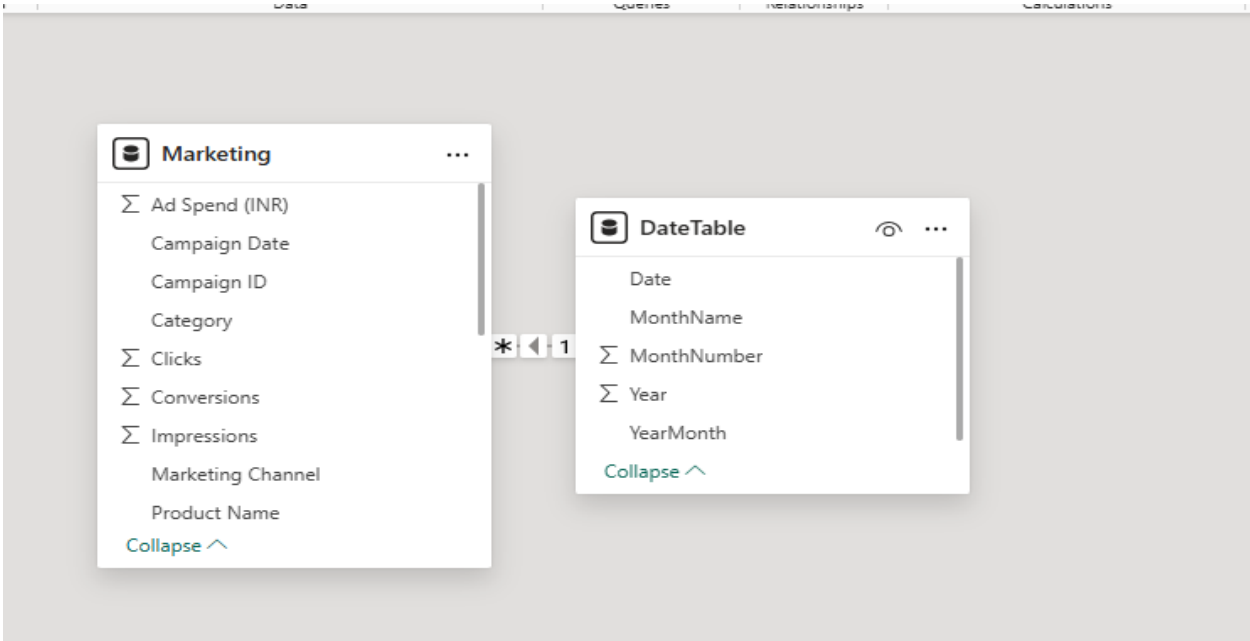
1. Data Preparation (in Power BI Desktop)

- **Imported the dataset** from PBI_Marketing_Data.xlsx.
- Verified data structure, data types (e.g., dates, numbers), and column names.
- Renamed fields for clarity (e.g., Ad_Spend to Ad Spend, Conv to Conversions).

2. Data Modeling

Created a clean **data model** with relationships where necessary.

Created a Date Table for time intelligence and established relationships with the Marketing table.



3. Created DAX Measures

- **Click-Through Rate (CTR):**

CTR = DIVIDE([Total_Clicks], [Total_Impressions], 0)

```
1 CTR = DIVIDE([Total_Clicks], [Total_Impressions], 0)
```

- **Conversion Rate:**

Conversion_Rate = DIVID([Total_Conversions], [Total_Clicks], 0)

```
1 Conversion_Rate = DIVIDE([Total_Conversions], [Total_Clicks], 0)
```

- **Return on Investment (ROI):**

ROI = DIVIDE([Total_Revenue] - [Total_Ad_ Spend]), [Total_Ad_ Spend], 0)

- Verified the calculated ROI against the existing column to ensure accuracy.

```
1 ROI_Calc = DIVIDE([Total_Revenue] - [Total_Ad_Spend], [Total_Ad_Spend], 0)
```

4. Key Performance Indicators (KPIs)

Used **Card Visuals** to show total metrics:

- Total Ad Spend

```
1 Total_Ad_Spend = SUM(marketing[Ad Spend (INR)])
```

- Total Impressions

```
1 Total_Impressions = SUM(marketing[Impressions])
```

- Total Clicks

```
1 Total_Clicks = SUM(marketing[Clicks])
```

- Total Conversions

```
1 Total_Conversions = SUM(marketing[Conversions])
```

- Total Revenue

```
1 Total_Revenue = SUM(marketing[Revenue (INR)])
```

- Overall ROI (as percentage)

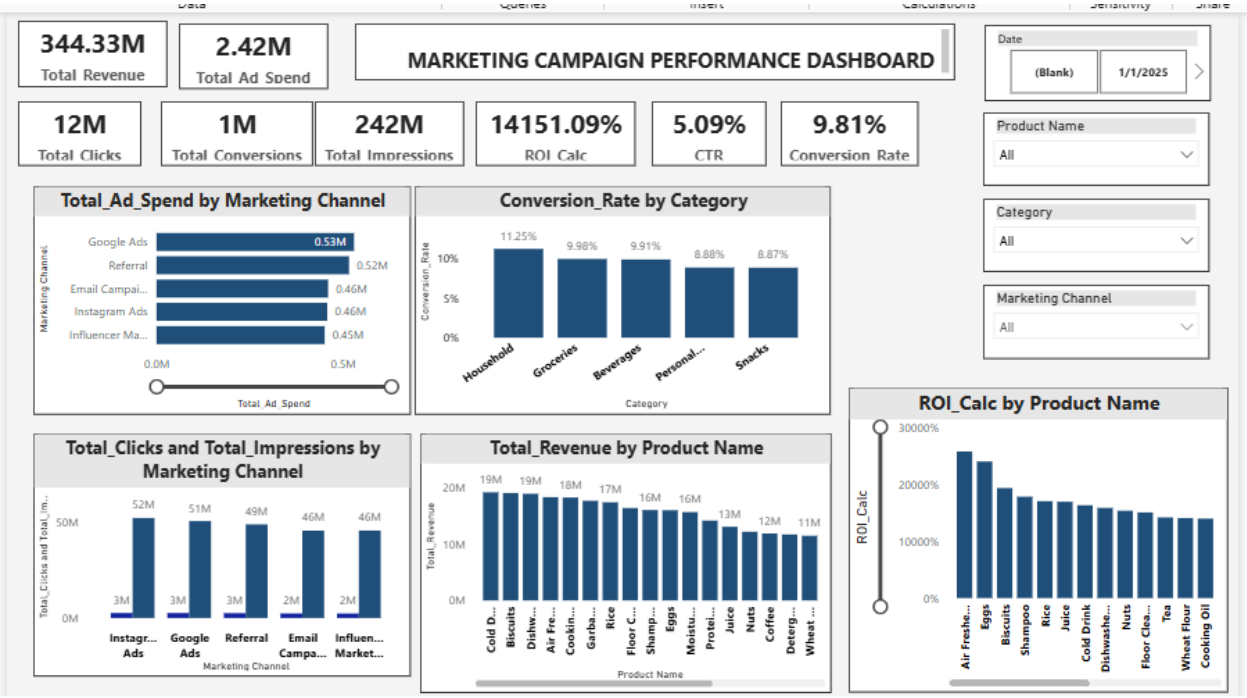
```
1 ROI_Calc = DIVIDE([Total_Revenue] - [Total_Ad_Spend], [Total_Ad_Spend], 0)
```

5. Visualizations

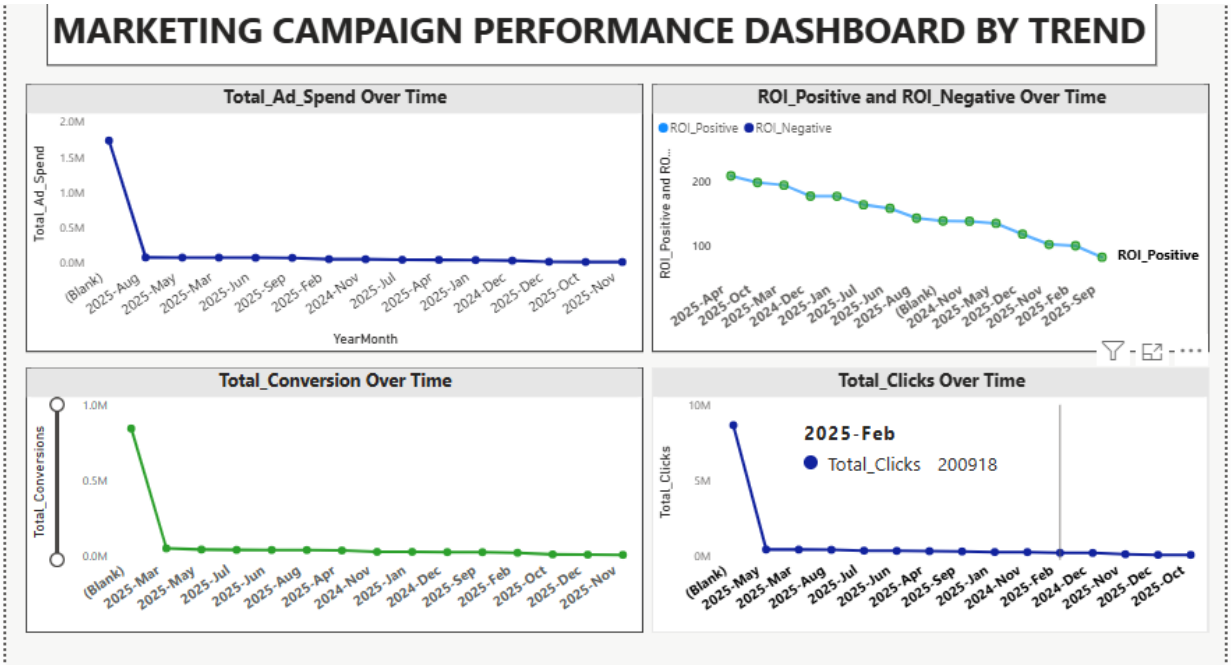
Designed with clarity and business users in mind:

- Ad Spend by Marketing Channel (Bar Chart): Comparison of budget allocation.
- Clicks vs Impressions by Channel (Clustered Column Chart): Channel performance in driving engagement.
- Conversion Rate by Category (Clustered Column Chart): Effectiveness across categories.
- Revenue by Product (Clustered Column Chart): Top revenue-generating products.
- ROI Trend Over Time (Line Chart) : Performance patterns across months.

This dashboard serves as a decision-support tool for the marketing team, enabling data-driven choices that align strategy with measurable performance outcomes.



- **Time-Based Trends** – Line Charts for:
  - Ad Spend over time
  - Conversions over time
  - ROI over time
  - Clicks over time



6. Interactivity Features

Added **Slicers** to enable dynamic filtering:

- **Campaign Date** – Date range slicer
- **Product Name** – Dropdown slicer
- **Product Category** – Dropdown slicer
- **Marketing Channel** – Dropdown slicer

All visuals respond dynamically to selected slicers, ensuring customized insights for the user.

7. Dashboard Design & Layout

- Applied a clean, minimalist layout with:
    - Right pane for filters/slicers
    - Central pane for visuals
    - Top row for KPIs
- Used **consistent color coding**:
    - Green for high ROI
    - Blue and grey tones for neutral visuals
- Added clear **dashboard title**.

Key Findings

1. ROI: Some channels consistently deliver positive ROI, while others perform below expectations.
2. Ad Spend Allocation: A significant portion of the budget is spent on channels that do not yield proportional revenue.
3. Clicks vs Impressions: Certain channels achieve high impressions but low clicks, indicating weak engagement.
4. Conversion Rate: Categories such as Electronics and Beauty have stronger conversion rates compared to Clothing.
5. Revenue Distribution: A few products (e.g., premium items) account for the majority of revenue.
6. Seasonality: Revenue and ROI trend upwards in Q4, suggesting holiday-season effectiveness.

Conclusion

The Power BI analysis highlighted inefficiencies in marketing spend, revealed top-performing channels and products, and confirmed the importance of seasonality. By reallocating budgets, focusing on high-performing categories, and improving weak engagement channels, the business can significantly increase overall ROI.

Recommendations

1. Optimize Ad Spend: Reallocate budget from low-performing channels to those with proven ROI.
2. Improve Engagement: Channels with high impressions but low clicks should have their messaging and creatives revised.
3. Focus on High-Converting Categories: Increase promotions and stock for categories with strong conversion rates.
4. Leverage Top Products: Maintain strong inventory and targeted campaigns for best-selling products.
5. Seasonal Strategy: Boost marketing spend during Q4 to maximize holiday sales opportunities.
6. Performance Monitoring: Continuously use the Power BI dashboard to monitor campaigns in real-time and adjust strategies.

Retail Sales SQL Project Report

1.0 Project Overview

This project demonstrates SQL data analysis skills using MySQL. The dataset represents retail sales transactions. The objectives were to set up a database, clean the data, perform exploratory data analysis (EDA), and answer specific business-related questions using SQL queries.

Database Setup and Data Import

Database Setup

1. A database named retail_sales was created:

CREATE DATABASE retail_sales;

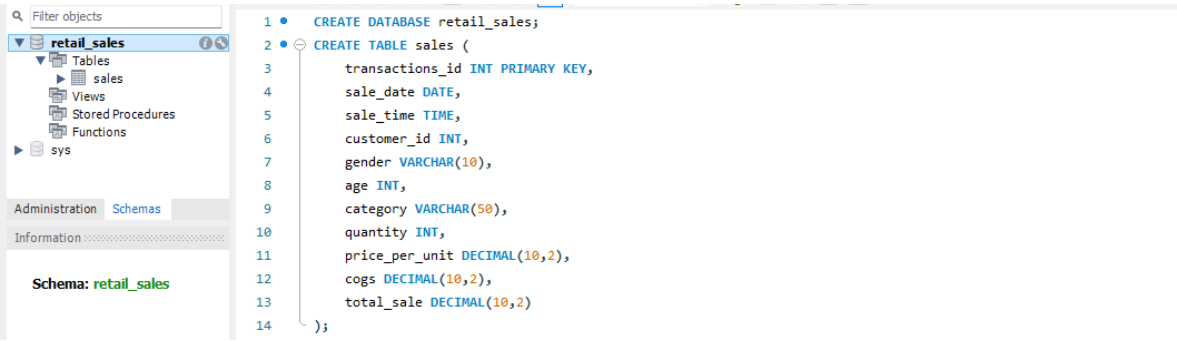
A sales table was created by importing the provided Excel dataset into MySQL Workbench. Data types were defined as follows:

CREATE TABLE Sales

- **transaction_id – INT (Primary Key)**
- **sale_date – DATE**
- **customer_id – INT**
- **gender – VARCHAR(10)**
- **category – VARCHAR(50)**
- **quantity – INT**
- **price – DECIMAL(10,2)**
- **payment_method – VARCHAR(50)**
- **total_sale – DECIMAL(10,2)**

```
CREATE TABLE sales (
    transactions_id INT PRIMARY KEY,
    sale_date DATE,
    sale_time TIME,
    customer_id INT,
    gender VARCHAR(10),
    age INT,
```

```
    category VARCHAR(50),
    quantity INT,
    price_per_unit DECIMAL(10,2),
    cogs DECIMAL(10,2),
    total_sale DECIMAL(10,2)
);
```



Data Cleaning

- **Checked for null values:**

SELECT*

FROM sales

WHERE sale_date IS NULL

OR customer_id IS NULL

OR gender IS NULL

OR age IS NULL

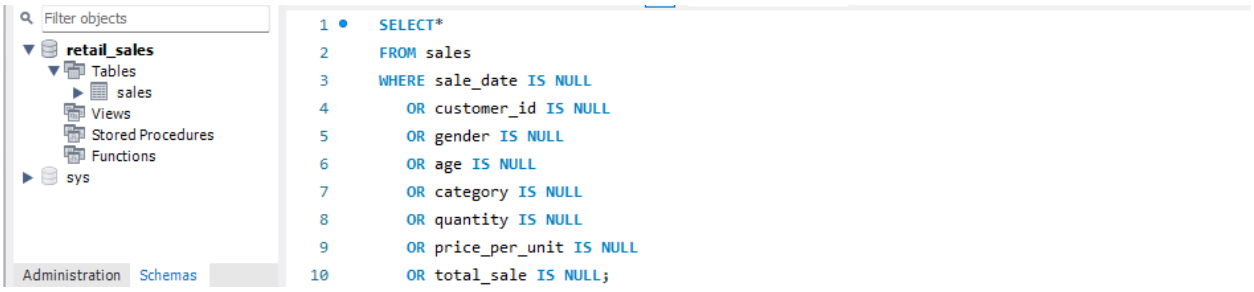
OR category IS NULL

OR quantity IS NULL

OR price_per_unit IS NULL

OR total_sale IS NULL;

- **No missing transaction_id values (Primary Key).**
- **Duplicates checked and removed if found.**



Deleted null values with:

DELETE FROM sales

WHERE sale_date IS NULL

OR customer_id IS NULL

OR gender IS NULL

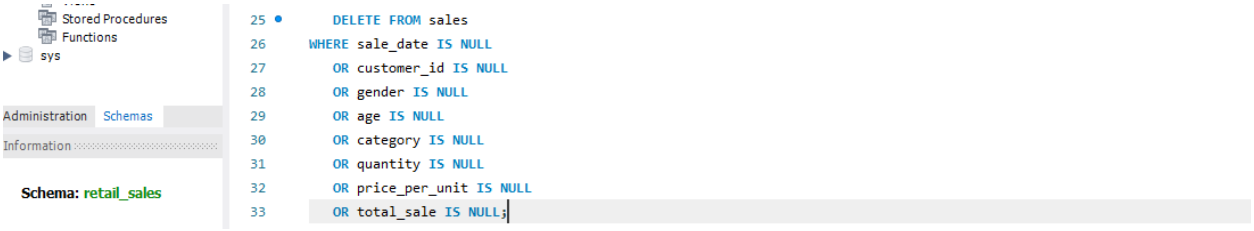
OR age IS NULL

OR category IS NULL

OR quantity IS NULL

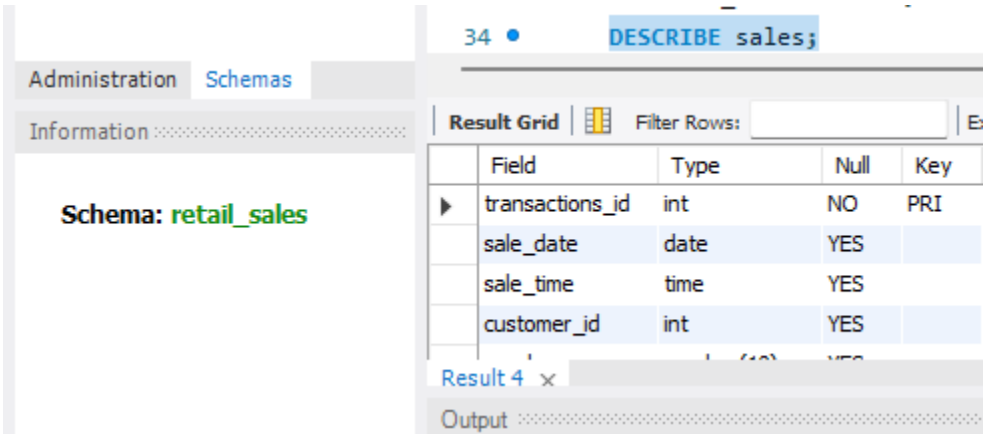
OR price_per_unit IS NULL

OR total_sale IS NULL;



- **Data types confirmed with:**

DESCRIBE sales;

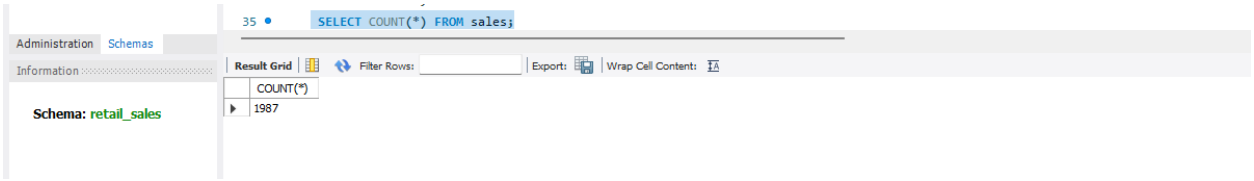


3. Exploratory Data Analysis (EDA)

Sample Queries:

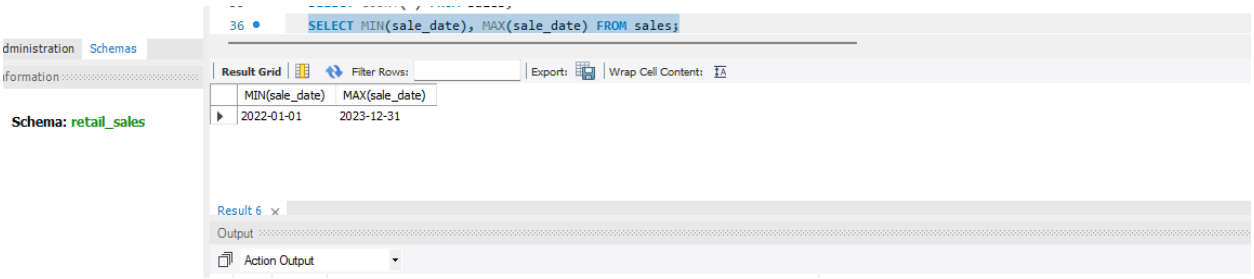
Total number of records

SELECT COUNT(*) FROM SalesData;



Range of sales dates:

SELECT MIN(sale_date), MAX(sale_date) FROM sales;

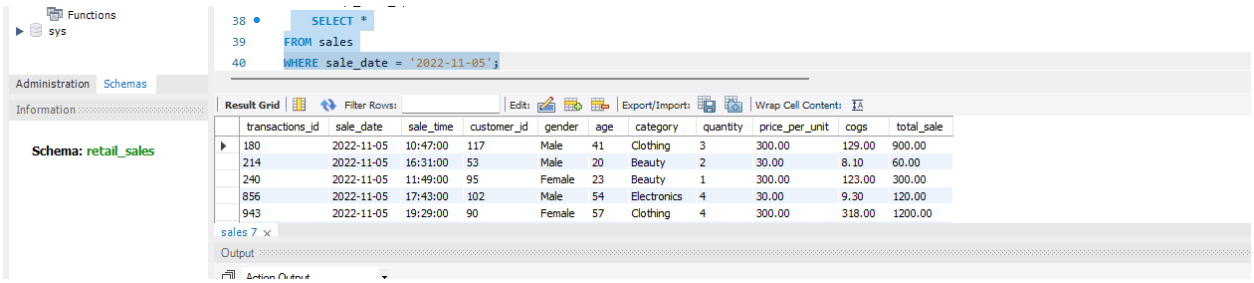


4. Business Questions & Solutions

i. Sales on 2022-11-05

SELECT * FROM Sales

WHERE date = '2022-11-05';



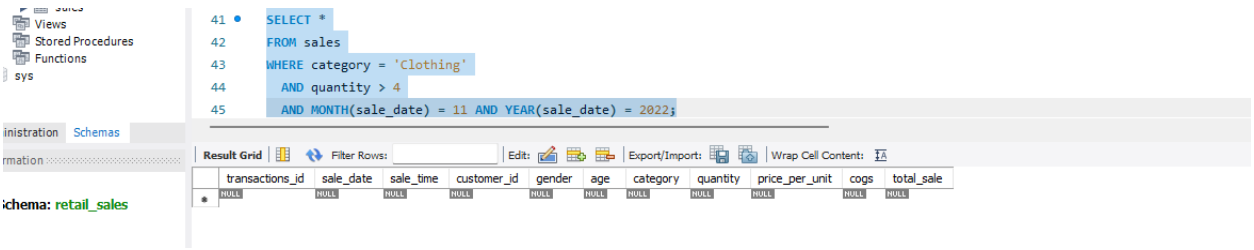
RESULT DISPLAYED ON EXCEL

	A	B	C	D	E	F	G	H	I	J	K
1	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
2	180	11/5/2022	10:47:00	117	Male	41	Clothing	3	300	129	900
3	214	11/5/2022	16:31:00	53	Male	20	Beauty	2	30	8.1	60
4	240	11/5/2022	11:49:00	95	Female	23	Beauty	1	300	123	300
5	856	11/5/2022	17:43:00	102	Male	54	Electronic	4	30	9.3	120
6	943	11/5/2022	19:29:00	90	Female	57	Clothing	4	300	318	1200
7	1137	11/5/2022	22:34:00	104	Male	46	Beauty	2	500	145	1000
8	1256	11/5/2022	9:58:00	29	Male	23	Clothing	2	500	190	1000
9	1265	11/5/2022	14:35:00	86	Male	55	Clothing	3	300	111	900
10	1587	11/5/2022	20:06:00	140	Female	40	Beauty	4	300	105	1200
11	1819	11/5/2022	20:44:00	83	Female	35	Beauty	2	50	13.5	100
12	1896	11/5/2022	20:19:00	87	Female	30	Electronic	2	25	30.75	50
13											

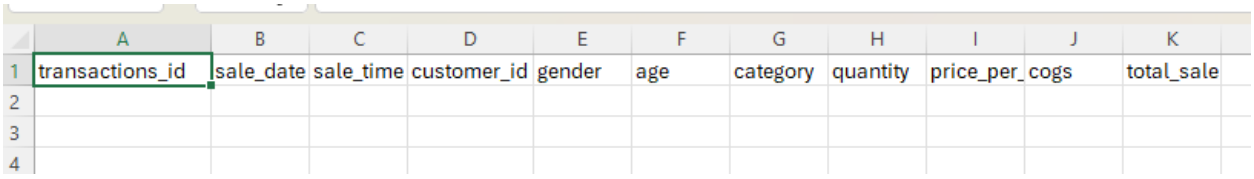
---

ii. Clothing category with quantity > 4 in Nov-2022

```
SELECT * FROM Sales
WHERE category = 'Clothing'
  AND quantity > 4
  AND MONTH(date) = 11
  AND YEAR(date) = 2022;
```

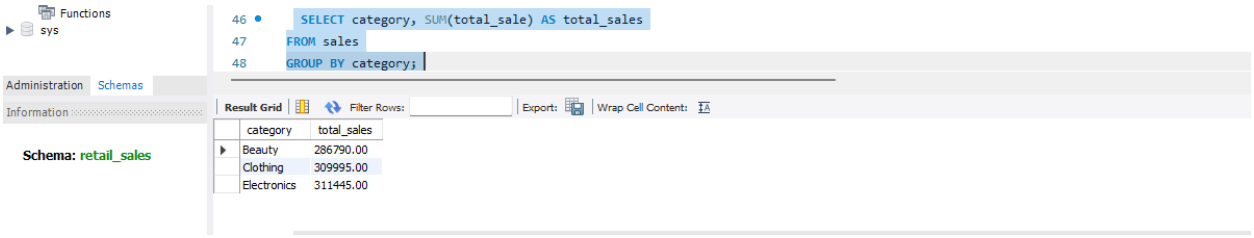


SCREENSHOT OF THE RESULT ON EXCEL

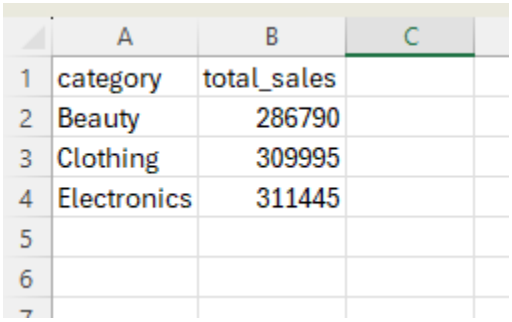


iii. Total sales per category

```
SELECT category, SUM(total_sale) AS total_sales
FROM Sales
GROUP BY category;
```



SCREENSHOT OF THE RESULT ON EXCEL



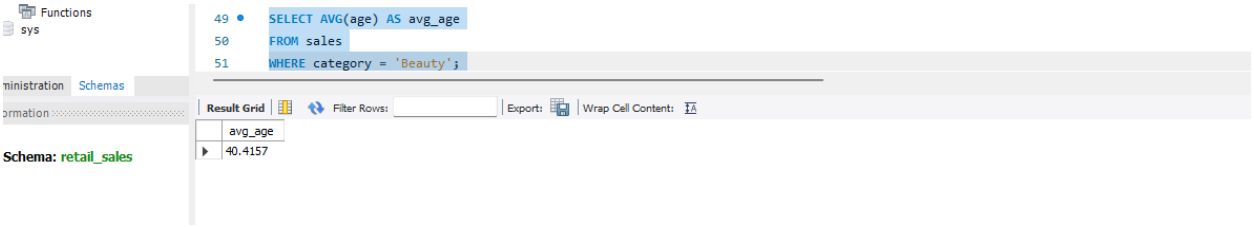
	A	B	C
1	category	total_sales	
2	Beauty	286790	
3	Clothing	309995	
4	Electronics	311445	
5			
6			

iv. Average age of customers who purchased Beauty items

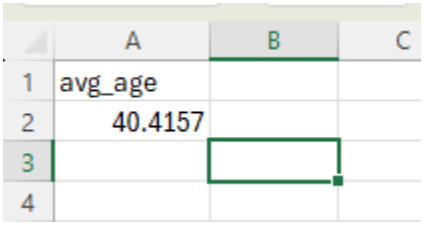
SELECT AVG(age) AS avg_age_beauty

FROM Sales

WHERE category = 'Beauty';



SCREENSHOT OF THE RESULT ON EXCEL

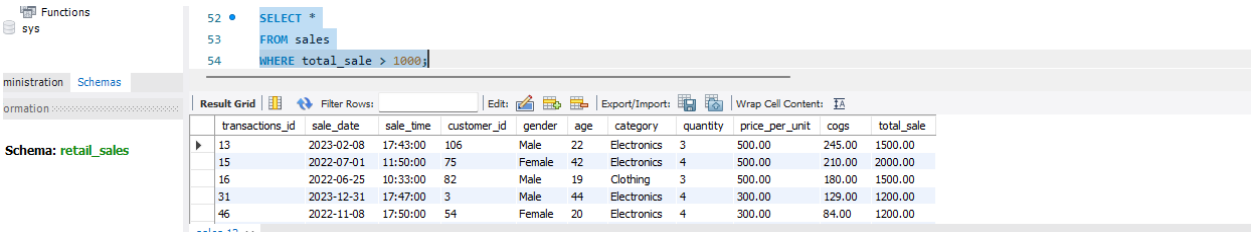


	A	B	C
1	avg_age		
2	40.4157		
3			
4			

v. Transactions with total_sale > 1000

SELECT * FROM Sales

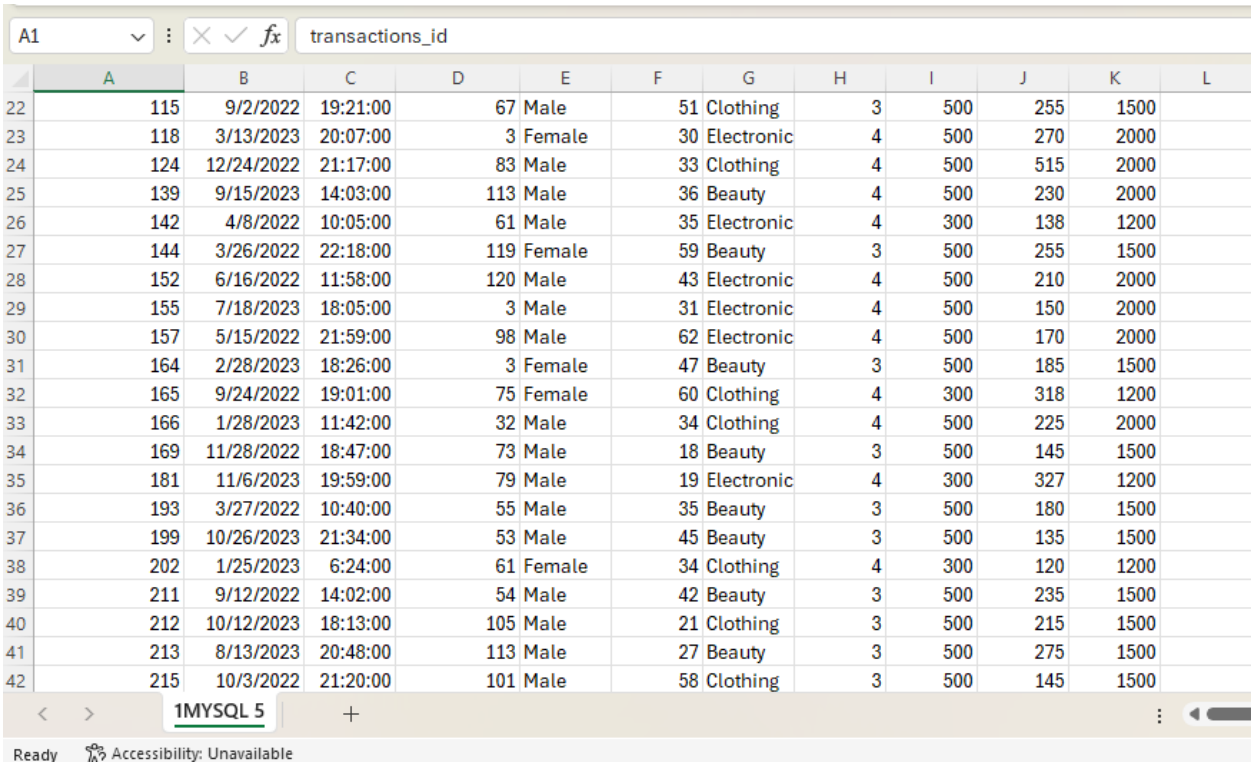
WHERE total_sale > 1000;



SCREENSHOT OF THE RESULT ON EXCEL

	A	B	C	D	E	F	G	H	I	J	K	L
1	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_	cogs	total_sale	
2	13	2/8/2023	17:43:00	106	Male	22	Electronic	3	500	245	1500	
3	15	7/1/2022	11:50:00	75	Female	42	Electronic	4	500	210	2000	
4	16	6/25/2022	10:33:00	82	Male	19	Clothing	3	500	180	1500	
5	31	12/31/2023	17:47:00	3	Male	44	Electronic	4	300	129	1200	
6	46	11/8/2022	17:50:00	54	Female	20	Electronic	4	300	84	1200	
7	47	10/22/2022	17:22:00	96	Female	40	Beauty	3	500	600	1500	
8	54	10/20/2022	10:17:00	142	Female	38	Electronic	3	500	200	1500	
9	58	9/16/2023	19:18:00	53	Male	18	Clothing	4	300	75	1200	
10	65	12/11/2022	20:03:00	84	Male	51	Electronic	4	500	160	2000	
11	67	8/19/2023	20:19:00	119	Female	48	Beauty	4	300	129	1200	
12	72	12/6/2023	19:19:00	5	Female	20	Electronic	4	500	195	2000	
13	74	10/5/2023	19:50:00	56	Female	18	Beauty	4	500	205	2000	
14	78	2/17/2023	21:08:00	68	Female	47	Clothing	3	500	265	1500	
15	89	12/30/2023	21:15:00	117	Female	55	Electronic	4	500	590	2000	
16	93	1/25/2022	20:52:00	148	Female	35	Beauty	4	500	140	2000	
17	99	11/19/2023	15:12:00	71	Female	50	Electronic	4	300	132	1200	
18	107	10/6/2022	9:18:00	75	Female	21	Clothing	4	300	78	1200	
19	109	9/6/2023	19:57:00	94	Female	34	Electronic	4	500	560	2000	
20	111	4/15/2023	9:45:00	5	Female	34	Electronic	3	500	130	1500	
21	112	12/25/2023	18:44:00	57	Male	37	Clothing	3	500	165	1500	

1MYSQL 5



A1 transcriptions_id

	A	B	C	D	E	F	G	H	I	J	K	L
22	115	9/2/2022	19:21:00	67	Male	51	Clothing	3	500	255	1500	
23	118	3/13/2023	20:07:00	3	Female	30	Electronic	4	500	270	2000	
24	124	12/24/2022	21:17:00	83	Male	33	Clothing	4	500	515	2000	
25	139	9/15/2023	14:03:00	113	Male	36	Beauty	4	500	230	2000	
26	142	4/8/2022	10:05:00	61	Male	35	Electronic	4	300	138	1200	
27	144	3/26/2022	22:18:00	119	Female	59	Beauty	3	500	255	1500	
28	152	6/16/2022	11:58:00	120	Male	43	Electronic	4	500	210	2000	
29	155	7/18/2023	18:05:00	3	Male	31	Electronic	4	500	150	2000	
30	157	5/15/2022	21:59:00	98	Male	62	Electronic	4	500	170	2000	
31	164	2/28/2023	18:26:00	3	Female	47	Beauty	3	500	185	1500	
32	165	9/24/2022	19:01:00	75	Female	60	Clothing	4	300	318	1200	
33	166	1/28/2023	11:42:00	32	Male	34	Clothing	4	500	225	2000	
34	169	11/28/2022	18:47:00	73	Male	18	Beauty	3	500	145	1500	
35	181	11/6/2023	19:59:00	79	Male	19	Electronic	4	300	327	1200	
36	193	3/27/2022	10:40:00	55	Male	35	Beauty	3	500	180	1500	
37	199	10/26/2023	21:34:00	53	Male	45	Beauty	3	500	135	1500	
38	202	1/25/2023	6:24:00	61	Female	34	Clothing	4	300	120	1200	
39	211	9/12/2022	14:02:00	54	Male	42	Beauty	3	500	235	1500	
40	212	10/12/2023	18:13:00	105	Male	21	Clothing	3	500	215	1500	
41	213	8/13/2023	20:48:00	113	Male	27	Beauty	3	500	275	1500	
42	215	10/3/2022	21:20:00	101	Male	58	Clothing	3	500	145	1500	

1MYSQL 5

Ready Accessibility: Unavailable

	A	B	C	D	E	F	G	H	I	J	K	L
43	239	9/19/2022	19:34:00	1	Male	38	Electronic	3	500	165	1500	
44	253	9/30/2022	21:26:00	66	Female	53	Clothing	4	500	525	2000	
45	257	12/10/2022	8:49:00	130	Male	19	Beauty	4	500	165	2000	
46	269	9/19/2022	11:31:00	87	Male	25	Clothing	4	500	250	2000	
47	280	11/17/2023	9:35:00	76	Female	37	Clothing	3	500	240	1500	
48	281	9/18/2023	20:09:00	4	Female	29	Beauty	4	500	275	2000	
49	292	10/10/2022	6:33:00	111	Male	20	Beauty	4	300	141	1200	
50	296	12/31/2022	22:44:00	73	Female	22	Clothing	4	300	369	1200	
51	298	10/5/2023	19:05:00	3	Male	27	Beauty	4	300	84	1200	
52	308	3/31/2022	19:12:00	130	Female	34	Beauty	4	300	126	1200	
53	313	12/16/2023	10:03:00	11	Female	55	Beauty	3	500	140	1500	
54	320	4/20/2022	8:35:00	57	Female	28	Electronic	4	300	159	1200	
55	332	4/24/2023	11:28:00	138	Male	58	Electronic	4	300	105	1200	
56	333	10/6/2022	8:15:00	21	Female	54	Electronic	4	300	99	1200	
57	340	9/19/2023	22:59:00	54	Female	36	Clothing	4	300	294	1200	
58	342	11/5/2023	17:24:00	85	Female	43	Clothing	4	500	500	2000	
59	356	8/27/2023	18:35:00	71	Male	50	Electronic	3	500	255	1500	
60	361	9/12/2022	16:37:00	112	Female	34	Electronic	4	300	81	1200	
61	368	1/31/2023	21:11:00	90	Female	56	Clothing	4	300	132	1200	
62	369	12/13/2023	19:42:00	111	Male	23	Electronic	3	500	215	1500	
63	372	6/19/2022	11:45:00	27	Female	24	Beauty	3	500	140	1500	

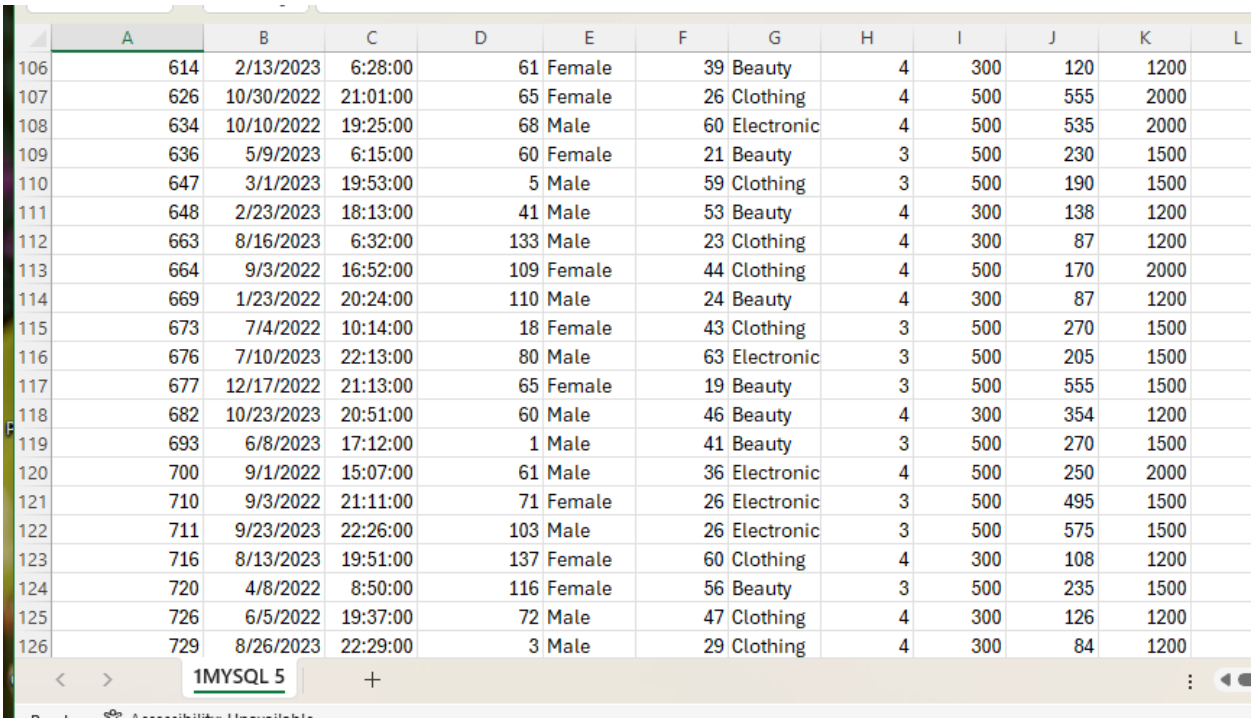
1MYSQL 5

	A	B	C	D	E	F	G	H	I	J	K	L
64	385	11/20/2022	19:06:00	127	Male	50	Electronic	3	500	525	1500	
65	405	10/17/2022	17:55:00	79	Female	25	Clothing	4	300	324	1200	
66	412	9/4/2022	19:36:00	30	Female	19	Electronic	4	500	570	2000	
67	416	8/21/2022	9:29:00	55	Male	53	Electronic	4	500	245	2000	
68	420	1/2/2022	10:53:00	28	Female	22	Clothing	4	500	200	2000	
69	421	4/8/2022	8:43:00	66	Female	37	Clothing	3	500	235	1500	
70	424	12/20/2023	19:21:00	101	Male	57	Beauty	4	300	147	1200	
71	431	10/29/2022	22:02:00	99	Male	63	Electronic	4	300	345	1200	
72	437	11/26/2023	18:53:00	56	Female	35	Electronic	4	300	345	1200	
73	441	11/6/2023	22:00:00	85	Male	57	Beauty	4	300	351	1200	
74	447	4/18/2023	17:57:00	37	Male	22	Beauty	4	500	245	2000	
75	452	1/30/2023	22:41:00	4	Female	48	Clothing	3	500	240	1500	
76	459	5/23/2023	7:12:00	122	Male	28	Clothing	4	300	75	1200	
77	462	8/26/2023	10:04:00	71	Male	63	Electronic	4	300	132	1200	
78	463	1/29/2023	20:22:00	55	Female	54	Beauty	3	500	175	1500	
79	474	5/8/2022	17:57:00	145	Female	26	Clothing	3	500	210	1500	
80	476	12/12/2023	18:31:00	131	Female	27	Clothing	4	500	570	2000	
81	479	11/3/2022	21:35:00	90	Male	52	Electronic	4	300	105	1200	
82	480	12/22/2022	22:12:00	126	Female	42	Beauty	4	500	225	2000	
83	481	7/13/2023	17:12:00	48	Female	43	Electronic	4	300	114	1200	
84	482	8/12/2023	18:57:00	3	Female	28	Clothing	4	300	141	1200	

1MYSQL 5

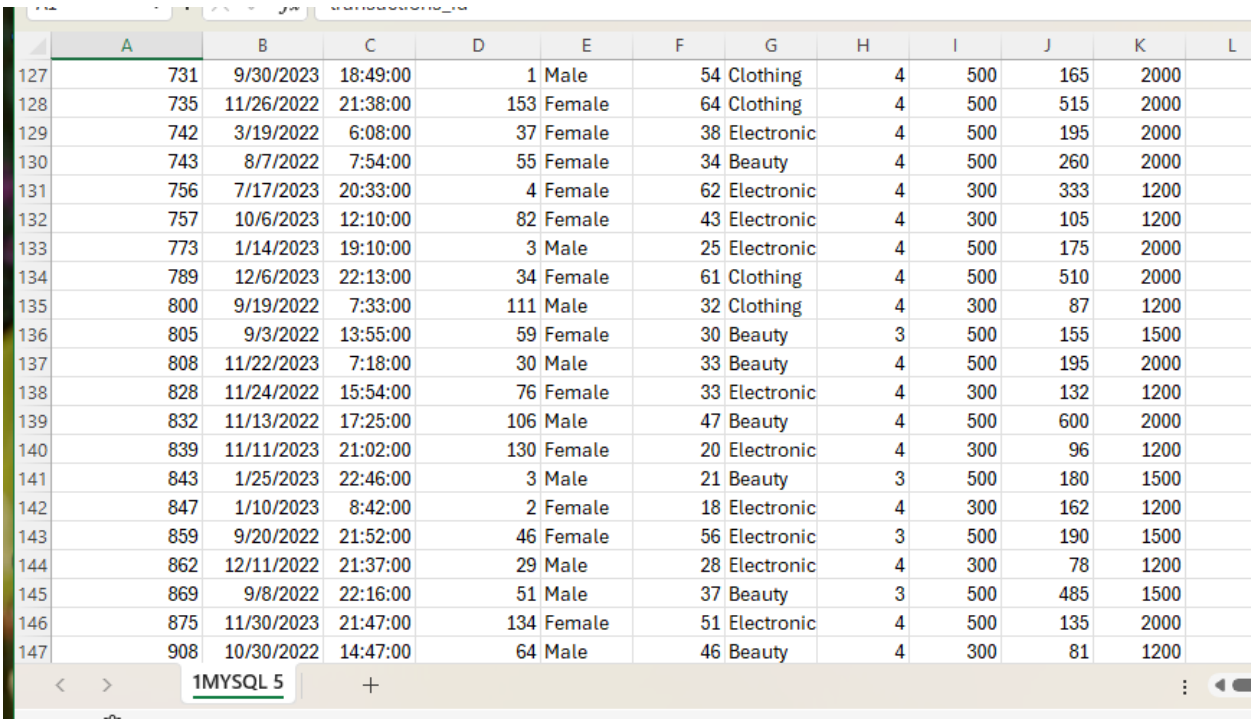
	A	B	C	D	E	F	G	H	I	J	K
85	484	3/13/2022	7:52:00	135	Female	19	Clothing	4	300	75	1200
86	487	4/9/2023	21:03:00	36	Male	44	Clothing	4	500	235	2000
87	503	10/8/2023	20:06:00	63	Male	45	Beauty	4	500	570	2000
88	506	4/29/2022	7:13:00	9	Male	34	Beauty	3	500	200	1500
89	507	12/11/2022	21:40:00	52	Female	37	Electronic	3	500	485	1500
90	522	7/9/2022	11:00:00	52	Male	46	Beauty	3	500	145	1500
91	524	9/17/2023	19:52:00	12	Male	46	Beauty	4	300	297	1200
92	533	11/13/2022	17:19:00	92	Male	19	Electronic	3	500	155	1500
93	547	11/14/2022	7:36:00	3	Male	63	Clothing	4	500	250	2000
94	553	3/20/2023	6:17:00	106	Male	24	Clothing	4	300	81	1200
95	559	12/12/2022	10:48:00	5	Female	40	Clothing	4	300	84	1200
96	561	7/29/2022	17:04:00	34	Female	64	Clothing	4	500	250	2000
97	572	12/13/2023	18:33:00	5	Male	31	Clothing	4	500	185	2000
98	577	4/21/2022	11:55:00	45	Male	21	Beauty	4	500	215	2000
99	580	11/14/2022	14:44:00	104	Female	31	Clothing	3	500	200	1500
100	587	9/4/2023	19:00:00	91	Female	40	Beauty	4	300	99	1200
101	592	12/26/2022	9:15:00	77	Female	46	Beauty	4	500	275	2000
102	595	10/15/2022	21:35:00	65	Female	18	Clothing	4	500	210	2000
103	597	11/1/2022	19:29:00	84	Male	22	Beauty	4	300	111	1200
104	608	12/7/2022	22:19:00	112	Female	55	Electronic	3	500	200	1500
105	611	7/21/2022	8:13:00	11	Male	51	Beauty	3	500	180	1500

1MYSQL 5



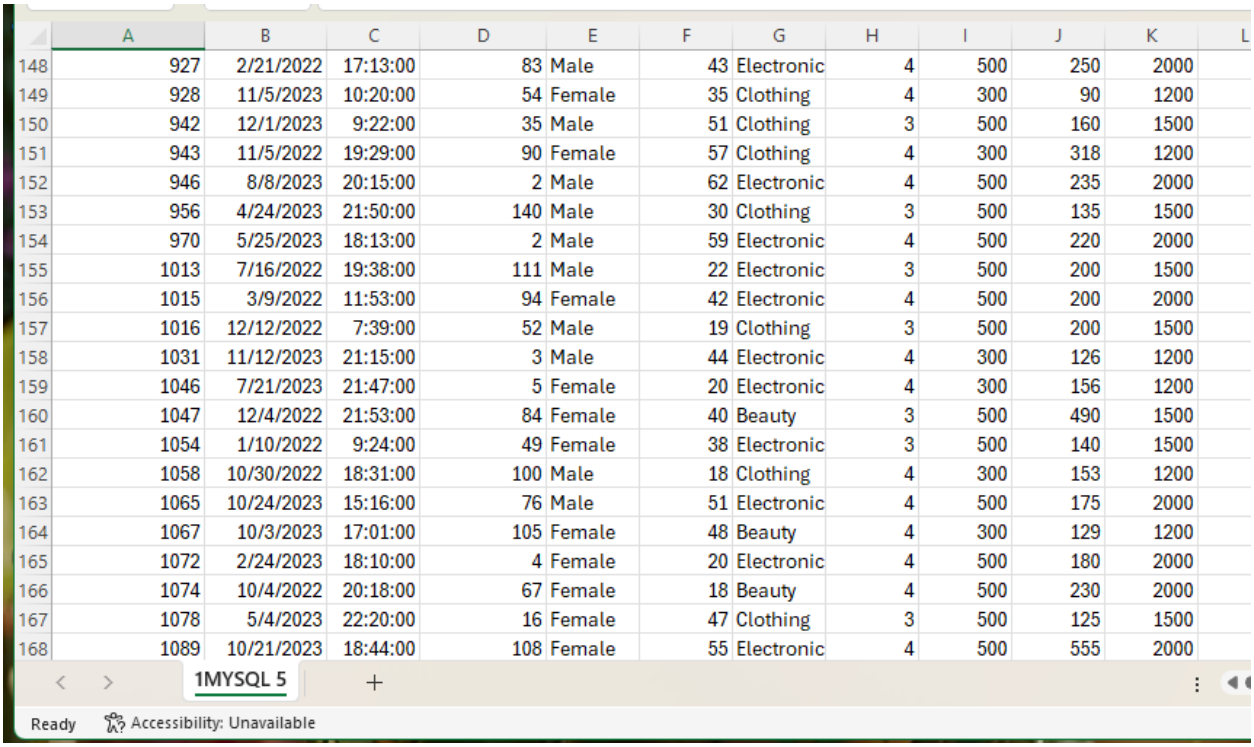
	A	B	C	D	E	F	G	H	I	J	K	L
106	614	2/13/2023	6:28:00	61	Female	39	Beauty	4	300	120	1200	
107	626	10/30/2022	21:01:00	65	Female	26	Clothing	4	500	555	2000	
108	634	10/10/2022	19:25:00	68	Male	60	Electronic	4	500	535	2000	
109	636	5/9/2023	6:15:00	60	Female	21	Beauty	3	500	230	1500	
110	647	3/1/2023	19:53:00	5	Male	59	Clothing	3	500	190	1500	
111	648	2/23/2023	18:13:00	41	Male	53	Beauty	4	300	138	1200	
112	663	8/16/2023	6:32:00	133	Male	23	Clothing	4	300	87	1200	
113	664	9/3/2022	16:52:00	109	Female	44	Clothing	4	500	170	2000	
114	669	1/23/2022	20:24:00	110	Male	24	Beauty	4	300	87	1200	
115	673	7/4/2022	10:14:00	18	Female	43	Clothing	3	500	270	1500	
116	676	7/10/2023	22:13:00	80	Male	63	Electronic	3	500	205	1500	
117	677	12/17/2022	21:13:00	65	Female	19	Beauty	3	500	555	1500	
118	682	10/23/2023	20:51:00	60	Male	46	Beauty	4	300	354	1200	
119	693	6/8/2023	17:12:00	1	Male	41	Beauty	3	500	270	1500	
120	700	9/1/2022	15:07:00	61	Male	36	Electronic	4	500	250	2000	
121	710	9/3/2022	21:11:00	71	Female	26	Electronic	3	500	495	1500	
122	711	9/23/2023	22:26:00	103	Male	26	Electronic	3	500	575	1500	
123	716	8/13/2023	19:51:00	137	Female	60	Clothing	4	300	108	1200	
124	720	4/8/2022	8:50:00	116	Female	56	Beauty	3	500	235	1500	
125	726	6/5/2022	19:37:00	72	Male	47	Clothing	4	300	126	1200	
126	729	8/26/2023	22:29:00	3	Male	29	Clothing	4	300	84	1200	

1MYSQL 5



	A	B	C	D	E	F	G	H	I	J	K	L
127	731	9/30/2023	18:49:00	1	Male	54	Clothing	4	500	165	2000	
128	735	11/26/2022	21:38:00	153	Female	64	Clothing	4	500	515	2000	
129	742	3/19/2022	6:08:00	37	Female	38	Electronic	4	500	195	2000	
130	743	8/7/2022	7:54:00	55	Female	34	Beauty	4	500	260	2000	
131	756	7/17/2023	20:33:00	4	Female	62	Electronic	4	300	333	1200	
132	757	10/6/2023	12:10:00	82	Female	43	Electronic	4	300	105	1200	
133	773	1/14/2023	19:10:00	3	Male	25	Electronic	4	500	175	2000	
134	789	12/6/2023	22:13:00	34	Female	61	Clothing	4	500	510	2000	
135	800	9/19/2022	7:33:00	111	Male	32	Clothing	4	300	87	1200	
136	805	9/3/2022	13:55:00	59	Female	30	Beauty	3	500	155	1500	
137	808	11/22/2023	7:18:00	30	Male	33	Beauty	4	500	195	2000	
138	828	11/24/2022	15:54:00	76	Female	33	Electronic	4	300	132	1200	
139	832	11/13/2022	17:25:00	106	Male	47	Beauty	4	500	600	2000	
140	839	11/11/2023	21:02:00	130	Female	20	Electronic	4	300	96	1200	
141	843	1/25/2023	22:46:00	3	Male	21	Beauty	3	500	180	1500	
142	847	1/10/2023	8:42:00	2	Female	18	Electronic	4	300	162	1200	
143	859	9/20/2022	21:52:00	46	Female	56	Electronic	3	500	190	1500	
144	862	12/11/2022	21:37:00	29	Male	28	Electronic	4	300	78	1200	
145	869	9/8/2022	22:16:00	51	Male	37	Beauty	3	500	485	1500	
146	875	11/30/2023	21:47:00	134	Female	51	Electronic	4	500	135	2000	
147	908	10/30/2022	14:47:00	64	Male	46	Beauty	4	300	81	1200	

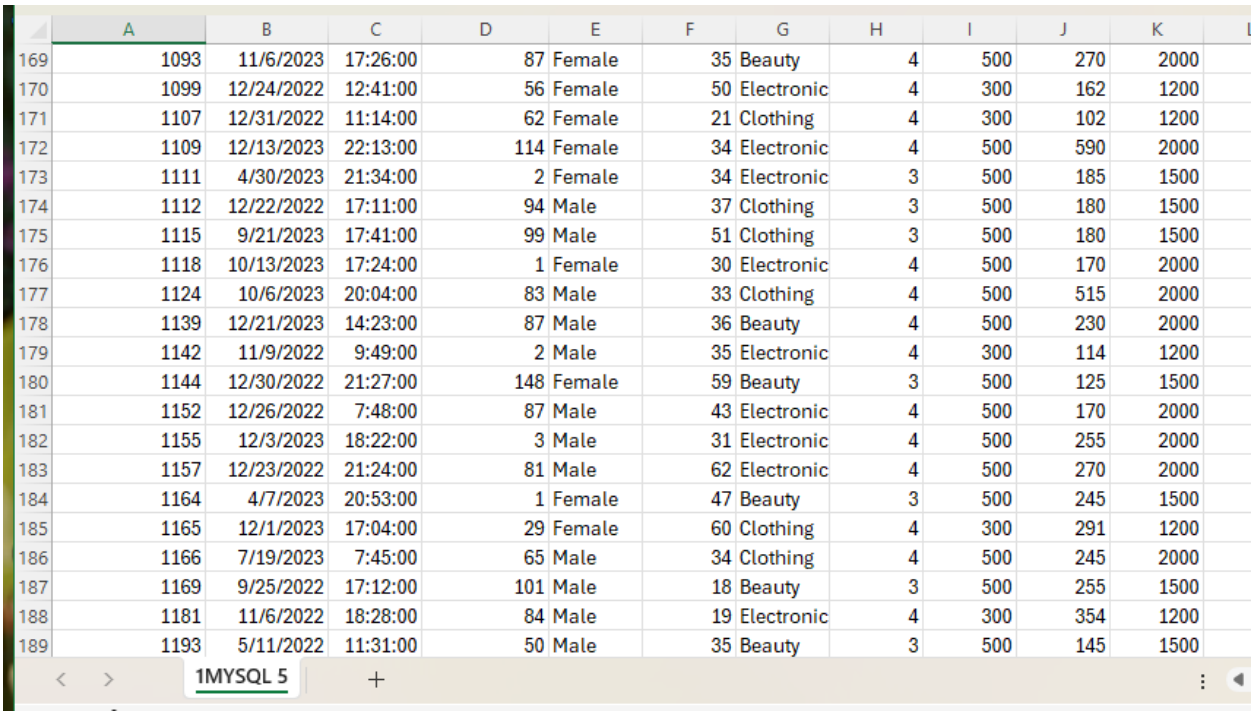
1MYSQL 5



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148	927	2/21/2022	17:13:00	83	Male	43	Electronic	4	500	250	2000
149	928	11/5/2023	10:20:00	54	Female	35	Clothing	4	300	90	1200
150	942	12/1/2023	9:22:00	35	Male	51	Clothing	3	500	160	1500
151	943	11/5/2022	19:29:00	90	Female	57	Clothing	4	300	318	1200
152	946	8/8/2023	20:15:00	2	Male	62	Electronic	4	500	235	2000
153	956	4/24/2023	21:50:00	140	Male	30	Clothing	3	500	135	1500
154	970	5/25/2023	18:13:00	2	Male	59	Electronic	4	500	220	2000
155	1013	7/16/2022	19:38:00	111	Male	22	Electronic	3	500	200	1500
156	1015	3/9/2022	11:53:00	94	Female	42	Electronic	4	500	200	2000
157	1016	12/12/2022	7:39:00	52	Male	19	Clothing	3	500	200	1500
158	1031	11/12/2023	21:15:00	3	Male	44	Electronic	4	300	126	1200
159	1046	7/21/2023	21:47:00	5	Female	20	Electronic	4	300	156	1200
160	1047	12/4/2022	21:53:00	84	Female	40	Beauty	3	500	490	1500
161	1054	1/10/2022	9:24:00	49	Female	38	Electronic	3	500	140	1500
162	1058	10/30/2022	18:31:00	100	Male	18	Clothing	4	300	153	1200
163	1065	10/24/2023	15:16:00	76	Male	51	Electronic	4	500	175	2000
164	1067	10/3/2023	17:01:00	105	Female	48	Beauty	4	300	129	1200
165	1072	2/24/2023	18:10:00	4	Female	20	Electronic	4	500	180	2000
166	1074	10/4/2022	20:18:00	67	Female	18	Beauty	4	500	230	2000
167	1078	5/4/2023	22:20:00	16	Female	47	Clothing	3	500	125	1500
168	1089	10/21/2023	18:44:00	108	Female	55	Electronic	4	500	555	2000

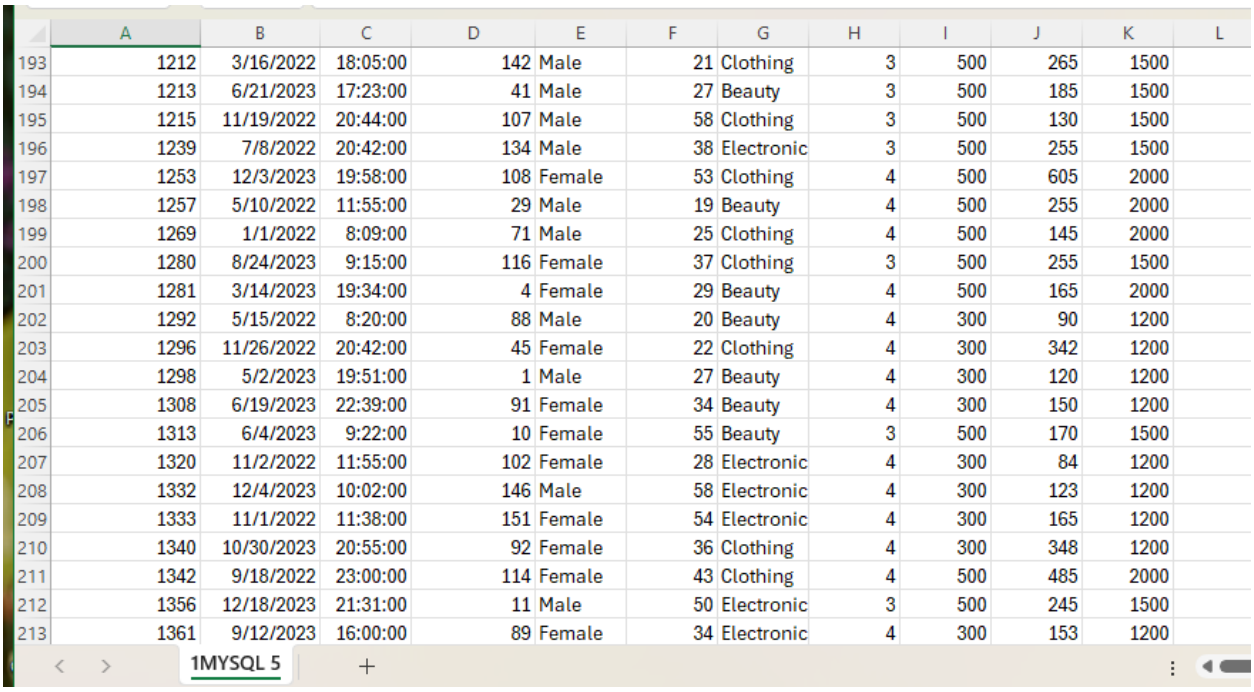
1MYSQL 5

Ready Accessibility: Unavailable



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169	1093	11/6/2023	17:26:00	87	Female	35	Beauty	4	500	270	2000
170	1099	12/24/2022	12:41:00	56	Female	50	Electronic	4	300	162	1200
171	1107	12/31/2022	11:14:00	62	Female	21	Clothing	4	300	102	1200
172	1109	12/13/2023	22:13:00	114	Female	34	Electronic	4	500	590	2000
173	1111	4/30/2023	21:34:00	2	Female	34	Electronic	3	500	185	1500
174	1112	12/22/2022	17:11:00	94	Male	37	Clothing	3	500	180	1500
175	1115	9/21/2023	17:41:00	99	Male	51	Clothing	3	500	180	1500
176	1118	10/13/2023	17:24:00	1	Female	30	Electronic	4	500	170	2000
177	1124	10/6/2023	20:04:00	83	Male	33	Clothing	4	500	515	2000
178	1139	12/21/2023	14:23:00	87	Male	36	Beauty	4	500	230	2000
179	1142	11/9/2022	9:49:00	2	Male	35	Electronic	4	300	114	1200
180	1144	12/30/2022	21:27:00	148	Female	59	Beauty	3	500	125	1500
181	1152	12/26/2022	7:48:00	87	Male	43	Electronic	4	500	170	2000
182	1155	12/3/2023	18:22:00	3	Male	31	Electronic	4	500	255	2000
183	1157	12/23/2022	21:24:00	81	Male	62	Electronic	4	500	270	2000
184	1164	4/7/2023	20:53:00	1	Female	47	Beauty	3	500	245	1500
185	1165	12/1/2023	17:04:00	29	Female	60	Clothing	4	300	291	1200
186	1166	7/19/2023	7:45:00	65	Male	34	Clothing	4	500	245	2000
187	1169	9/25/2022	17:12:00	101	Male	18	Beauty	3	500	255	1500
188	1181	11/6/2022	18:28:00	84	Male	19	Electronic	4	300	354	1200
189	1193	5/11/2022	11:31:00	50	Male	35	Beauty	3	500	145	1500

1MYSQL 5



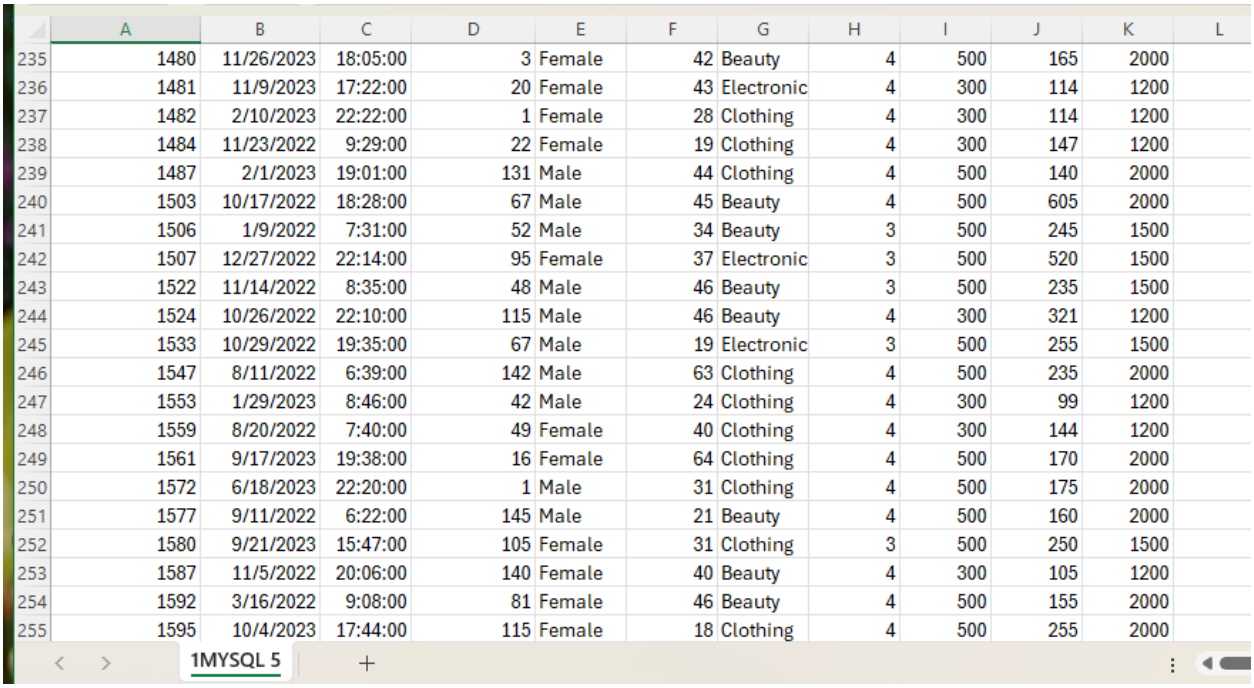
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193	1212	3/16/2022	18:05:00	142	Male	21	Clothing	3	500	265	1500	
194	1213	6/21/2023	17:23:00	41	Male	27	Beauty	3	500	185	1500	
195	1215	11/19/2022	20:44:00	107	Male	58	Clothing	3	500	130	1500	
196	1239	7/8/2022	20:42:00	134	Male	38	Electronic	3	500	255	1500	
197	1253	12/3/2023	19:58:00	108	Female	53	Clothing	4	500	605	2000	
198	1257	5/10/2022	11:55:00	29	Male	19	Beauty	4	500	255	2000	
199	1269	1/1/2022	8:09:00	71	Male	25	Clothing	4	500	145	2000	
200	1280	8/24/2023	9:15:00	116	Female	37	Clothing	3	500	255	1500	
201	1281	3/14/2023	19:34:00	4	Female	29	Beauty	4	500	165	2000	
202	1292	5/15/2022	8:20:00	88	Male	20	Beauty	4	300	90	1200	
203	1296	11/26/2022	20:42:00	45	Female	22	Clothing	4	300	342	1200	
204	1298	5/2/2023	19:51:00	1	Male	27	Beauty	4	300	120	1200	
205	1308	6/19/2023	22:39:00	91	Female	34	Beauty	4	300	150	1200	
206	1313	6/4/2023	9:22:00	10	Female	55	Beauty	3	500	170	1500	
207	1320	11/2/2022	11:55:00	102	Female	28	Electronic	4	300	84	1200	
208	1332	12/4/2023	10:02:00	146	Male	58	Electronic	4	300	123	1200	
209	1333	11/1/2022	11:38:00	151	Female	54	Electronic	4	300	165	1200	
210	1340	10/30/2023	20:55:00	92	Female	36	Clothing	4	300	348	1200	
211	1342	9/18/2022	23:00:00	114	Female	43	Clothing	4	500	485	2000	
212	1356	12/18/2023	21:31:00	11	Male	50	Electronic	3	500	245	1500	
213	1361	9/12/2023	16:00:00	89	Female	34	Electronic	4	300	153	1200	

1MYSQL 5



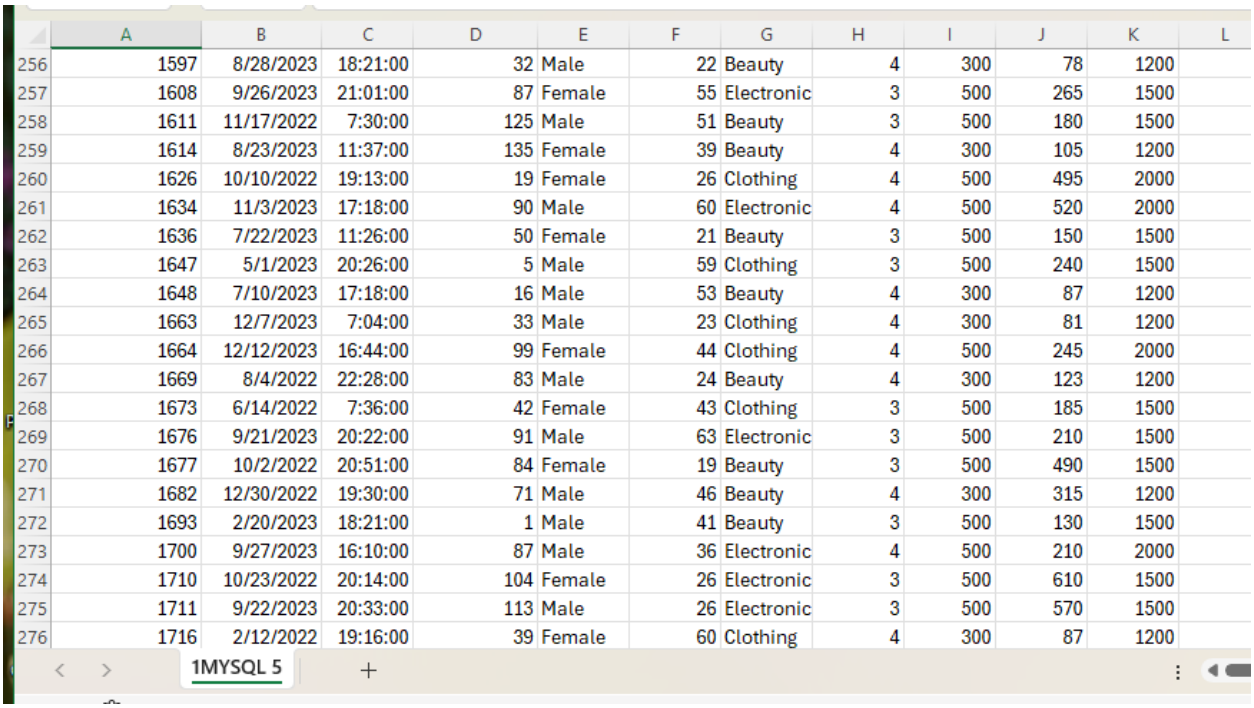
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214	1368	3/8/2022	21:29:00	145	Female	56	Clothing	4	300	132	1200	
215	1369	10/10/2022	21:14:00	64	Male	23	Electronic	3	500	180	1500	
216	1372	3/18/2022	10:53:00	60	Female	24	Beauty	3	500	145	1500	
217	1385	11/7/2023	19:22:00	55	Male	50	Electronic	3	500	505	1500	
218	1405	10/28/2023	20:08:00	91	Female	25	Clothing	4	300	342	1200	
219	1412	10/13/2022	19:59:00	75	Female	19	Electronic	4	500	615	2000	
220	1416	12/29/2022	10:47:00	111	Male	53	Electronic	4	500	125	2000	
221	1420	4/15/2022	7:01:00	138	Female	22	Clothing	4	500	205	2000	
222	1421	1/17/2022	7:07:00	59	Female	37	Clothing	3	500	185	1500	
223	1424	9/5/2023	21:32:00	54	Male	57	Beauty	4	300	111	1200	
224	1431	9/28/2023	21:28:00	114	Male	63	Electronic	4	300	312	1200	
225	1437	9/4/2023	19:19:00	76	Female	35	Electronic	4	300	372	1200	
226	1441	10/4/2023	21:38:00	63	Male	57	Beauty	4	300	354	1200	
227	1447	5/30/2022	22:21:00	137	Male	22	Beauty	4	500	155	2000	
228	1452	8/6/2023	17:41:00	3	Female	48	Clothing	3	500	185	1500	
229	1459	11/10/2023	6:46:00	13	Male	28	Clothing	4	300	144	1200	
230	1462	11/13/2023	10:42:00	124	Male	63	Electronic	4	300	114	1200	
231	1463	7/29/2022	19:45:00	98	Female	54	Beauty	3	500	170	1500	
232	1474	5/15/2022	20:49:00	84	Female	26	Clothing	3	500	255	1500	
233	1476	11/11/2022	22:27:00	130	Female	27	Clothing	4	500	555	2000	
234	1479	5/31/2022	22:24:00	108	Male	52	Electronic	4	300	162	1200	

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	A	B	C	D	E	F	G	H	I	J	K	L
235	1480	11/26/2023	18:05:00	3	Female	42	Beauty	4	500	165	2000	
236	1481	11/9/2023	17:22:00	20	Female	43	Electronic	4	300	114	1200	
237	1482	2/10/2023	22:22:00	1	Female	28	Clothing	4	300	114	1200	
238	1484	11/23/2022	9:29:00	22	Female	19	Clothing	4	300	147	1200	
239	1487	2/1/2023	19:01:00	131	Male	44	Clothing	4	500	140	2000	
240	1503	10/17/2022	18:28:00	67	Male	45	Beauty	4	500	605	2000	
241	1506	1/9/2022	7:31:00	52	Male	34	Beauty	3	500	245	1500	
242	1507	12/27/2022	22:14:00	95	Female	37	Electronic	3	500	520	1500	
243	1522	11/14/2022	8:35:00	48	Male	46	Beauty	3	500	235	1500	
244	1524	10/26/2022	22:10:00	115	Male	46	Beauty	4	300	321	1200	
245	1533	10/29/2022	19:35:00	67	Male	19	Electronic	3	500	255	1500	
246	1547	8/11/2022	6:39:00	142	Male	63	Clothing	4	500	235	2000	
247	1553	1/29/2023	8:46:00	42	Male	24	Clothing	4	300	99	1200	
248	1559	8/20/2022	7:40:00	49	Female	40	Clothing	4	300	144	1200	
249	1561	9/17/2023	19:38:00	16	Female	64	Clothing	4	500	170	2000	
250	1572	6/18/2023	22:20:00	1	Male	31	Clothing	4	500	175	2000	
251	1577	9/11/2022	6:22:00	145	Male	21	Beauty	4	500	160	2000	
252	1580	9/21/2023	15:47:00	105	Female	31	Clothing	3	500	250	1500	
253	1587	11/5/2022	20:06:00	140	Female	40	Beauty	4	300	105	1200	
254	1592	3/16/2022	9:08:00	81	Female	46	Beauty	4	500	155	2000	
255	1595	10/4/2023	17:44:00	115	Female	18	Clothing	4	500	255	2000	

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	A	B	C	D	E	F	G	H	I	J	K	L
256	1597	8/28/2023	18:21:00	32	Male	22	Beauty	4	300	78	1200	
257	1608	9/26/2023	21:01:00	87	Female	55	Electronic	3	500	265	1500	
258	1611	11/17/2022	7:30:00	125	Male	51	Beauty	3	500	180	1500	
259	1614	8/23/2023	11:37:00	135	Female	39	Beauty	4	300	105	1200	
260	1626	10/10/2022	19:13:00	19	Female	26	Clothing	4	500	495	2000	
261	1634	11/3/2023	17:18:00	90	Male	60	Electronic	4	500	520	2000	
262	1636	7/22/2023	11:26:00	50	Female	21	Beauty	3	500	150	1500	
263	1647	5/1/2023	20:26:00	5	Male	59	Clothing	3	500	240	1500	
264	1648	7/10/2023	17:18:00	16	Male	53	Beauty	4	300	87	1200	
265	1663	12/7/2023	7:04:00	33	Male	23	Clothing	4	300	81	1200	
266	1664	12/12/2023	16:44:00	99	Female	44	Clothing	4	500	245	2000	
267	1669	8/4/2022	22:28:00	83	Male	24	Beauty	4	300	123	1200	
268	1673	6/14/2022	7:36:00	42	Female	43	Clothing	3	500	185	1500	
269	1676	9/21/2023	20:22:00	91	Male	63	Electronic	3	500	210	1500	
270	1677	10/2/2022	20:51:00	84	Female	19	Beauty	3	500	490	1500	
271	1682	12/30/2022	19:30:00	71	Male	46	Beauty	4	300	315	1200	
272	1693	2/20/2023	18:21:00	1	Male	41	Beauty	3	500	130	1500	
273	1700	9/27/2023	16:10:00	87	Male	36	Electronic	4	500	210	2000	
274	1710	10/23/2022	20:14:00	104	Female	26	Electronic	3	500	610	1500	
275	1711	9/22/2023	20:33:00	113	Male	26	Electronic	3	500	570	1500	
276	1716	2/12/2022	19:16:00	39	Female	60	Clothing	4	300	87	1200	

1MYSQL 5

	A	B	C	D	E	F	G	H	I	J	K	L
277	1720	10/10/2022	8:51:00	28	Female	56	Beauty	3	500	190	1500	
278	1726	6/26/2023	18:12:00	13	Male	47	Clothing	4	300	93	1200	
279	1729	11/22/2023	18:55:00	82	Male	29	Clothing	4	300	81	1200	
280	1731	12/19/2023	17:41:00	3	Male	54	Clothing	4	500	195	2000	
281	1735	9/5/2023	17:26:00	86	Female	64	Clothing	4	500	505	2000	
282	1742	11/22/2022	8:25:00	18	Female	38	Electronic	4	500	220	2000	
283	1743	10/26/2022	9:37:00	47	Female	34	Beauty	4	500	250	2000	
284	1756	5/12/2022	22:16:00	119	Female	62	Electronic	4	300	297	1200	
285	1757	9/29/2023	16:42:00	50	Female	43	Electronic	4	300	132	1200	
286	1773	12/12/2023	18:50:00	80	Male	25	Electronic	4	500	165	2000	
287	1789	11/10/2023	17:02:00	144	Female	61	Clothing	4	500	615	2000	
288	1800	8/21/2022	10:53:00	1	Male	32	Clothing	4	300	111	1200	
289	1805	10/10/2023	13:35:00	79	Female	30	Beauty	3	500	225	1500	
290	1808	2/12/2023	10:48:00	109	Male	33	Beauty	4	500	210	2000	
291	1828	10/14/2022	14:14:00	99	Female	33	Electronic	4	300	165	1200	
292	1832	11/10/2023	21:17:00	57	Male	47	Beauty	4	500	480	2000	
293	1839	8/28/2022	21:57:00	92	Female	20	Electronic	4	300	156	1200	
294	1843	9/21/2023	18:58:00	1	Male	21	Beauty	3	500	240	1500	
295	1847	7/28/2023	7:14:00	5	Female	18	Electronic	4	300	123	1200	
296	1859	9/14/2022	20:58:00	32	Female	56	Electronic	3	500	205	1500	
297	1862	2/6/2022	22:50:00	64	Male	28	Electronic	4	300	165	1200	

1MYSQL 5

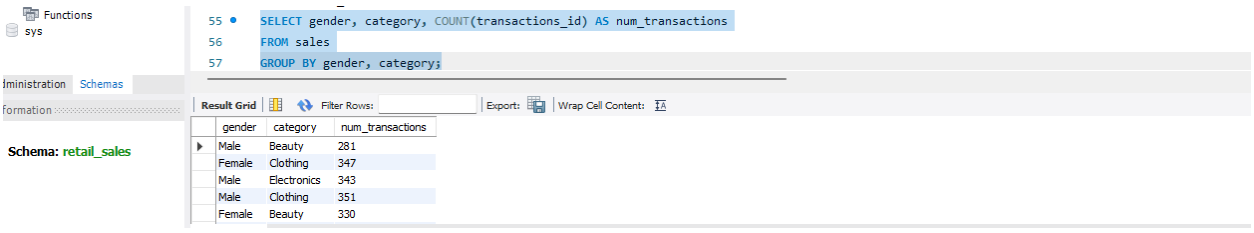
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298	1869	10/23/2022	19:51:00	113	Male	37	Beauty	3	500	605	1500	
299	1875	6/3/2022	22:42:00	141	Female	51	Electronic	4	500	165	2000	
300	1908	12/17/2023	12:34:00	93	Male	46	Beauty	4	300	87	1200	
301	1927	9/6/2023	19:44:00	46	Male	43	Electronic	4	500	180	2000	
302	1928	7/29/2023	6:39:00	93	Female	35	Clothing	4	300	75	1200	
303	1942	9/16/2023	11:53:00	112	Male	51	Clothing	3	500	265	1500	
304	1943	11/17/2023	21:00:00	108	Female	57	Clothing	4	300	321	1200	
305	1946	7/27/2023	20:48:00	5	Male	62	Electronic	4	500	165	2000	
306	1956	6/1/2023	20:40:00	62	Male	30	Clothing	3	500	170	1500	
307	1970	5/22/2023	17:44:00	5	Male	59	Electronic	4	500	230	2000	
308												

vi. Transactions by gender in each category

SELECT gender, category, COUNT(transaction_id) AS num_transactions

FROM Sales

GROUP BY gender, category;

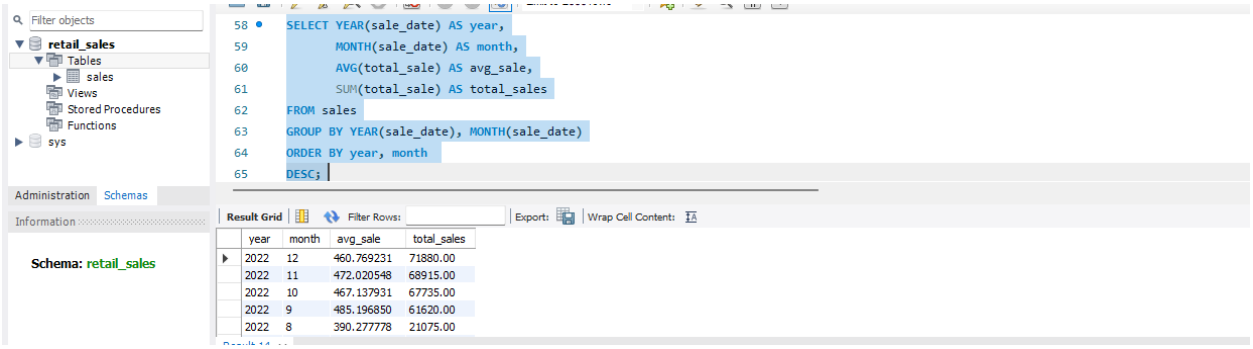


SCREENSHOT OF THE RESULT ON EXCEL

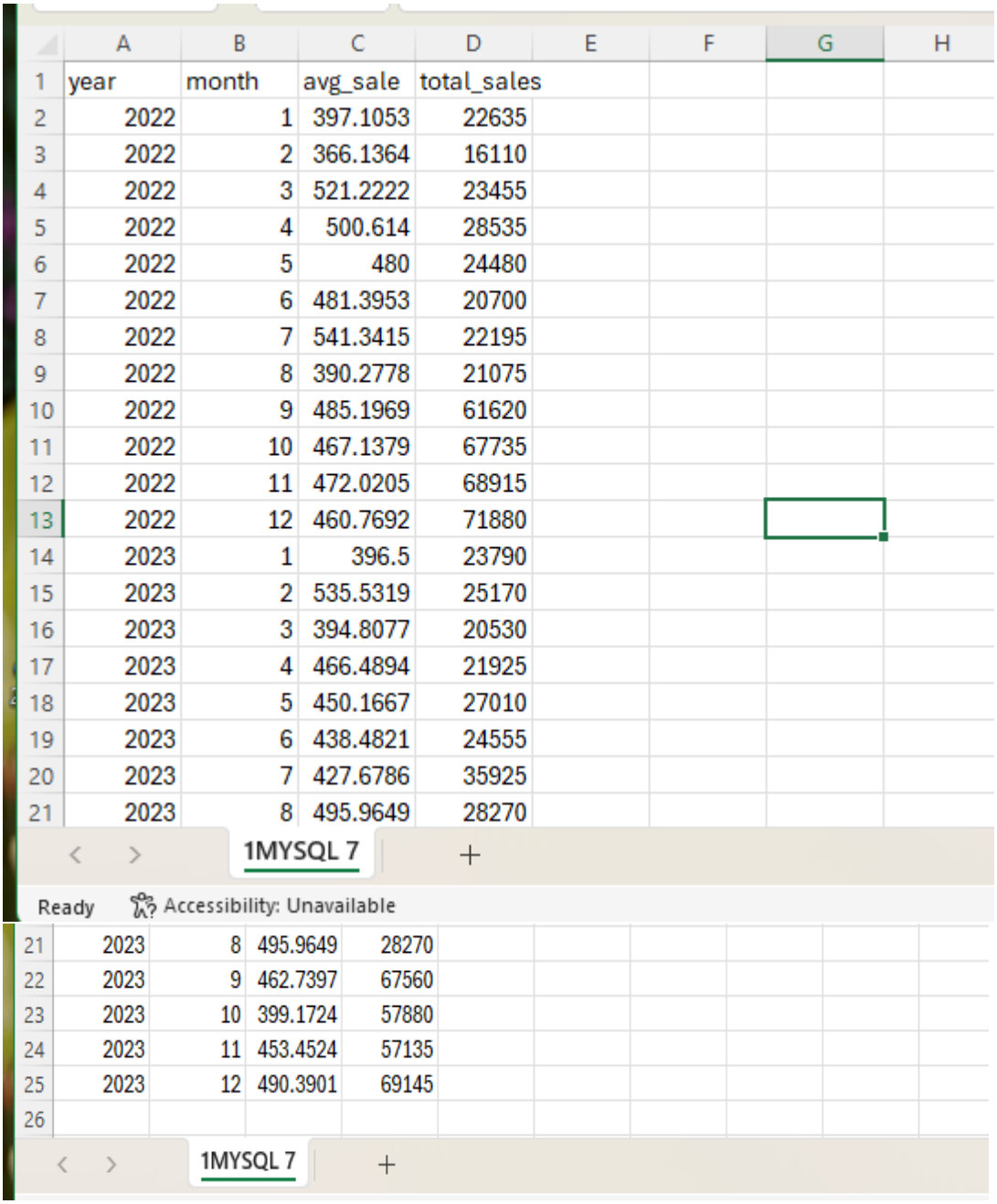
	A	B	C	D
1	gender	category	num_transactions	
2	Male	Beauty	281	
3	Female	Clothing	347	
4	Male	Electronics	343	
5	Male	Clothing	351	
6	Female	Beauty	330	
7	Female	Electronics	335	
8				
9				

vii. Average sale per month & best-selling month

```
SELECT YEAR(sale_date) AS year,
    MONTH(sale_date) AS month,
    AVG(total_sale) AS avg_sale,
    SUM(total_sale) AS total_sales
FROM sales
GROUP BY YEAR(sale_date), MONTH(sale_date)
ORDER BY year, month
DESC;
```



SCREENSHOT OF RESULT ON EXCEL

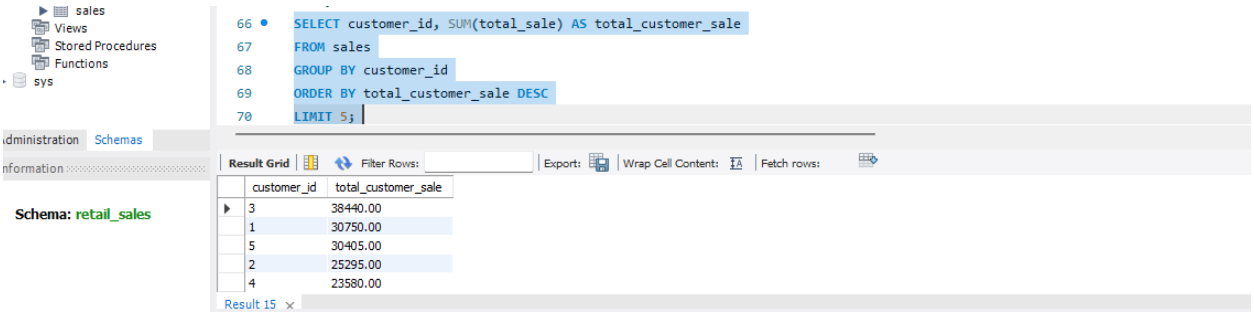


	year	month	avg_sale	total_sales
2	2022	1	397.1053	22635
3	2022	2	366.1364	16110
4	2022	3	521.2222	23455
5	2022	4	500.614	28535
6	2022	5	480	24480
7	2022	6	481.3953	20700
8	2022	7	541.3415	22195
9	2022	8	390.2778	21075
10	2022	9	485.1969	61620
11	2022	10	467.1379	67735
12	2022	11	472.0205	68915
13	2022	12	460.7692	71880
14	2023	1	396.5	23790
15	2023	2	535.5319	25170
16	2023	3	394.8077	20530
17	2023	4	466.4894	21925
18	2023	5	450.1667	27010
19	2023	6	438.4821	24555
20	2023	7	427.6786	35925
21	2023	8	495.9649	28270
22	2023	9	462.7397	67560
23	2023	10	399.1724	57880
24	2023	11	453.4524	57135
25	2023	12	490.3901	69145

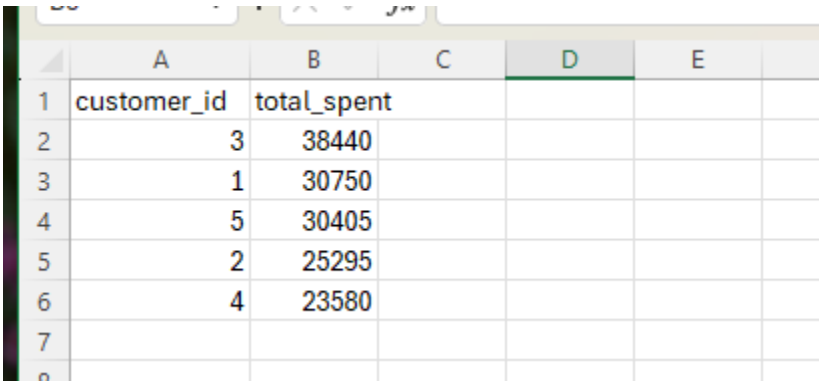
viii. Top 5 customers by total sales

```
SELECT customer_id, SUM(total_sale) AS total_customer_sales
FROM Sales
GROUP BY customer_id
ORDER BY total_customer_sales DESC
```

LIMIT 5;



SCREENSHOT OF THE RESULT ON EXCEL



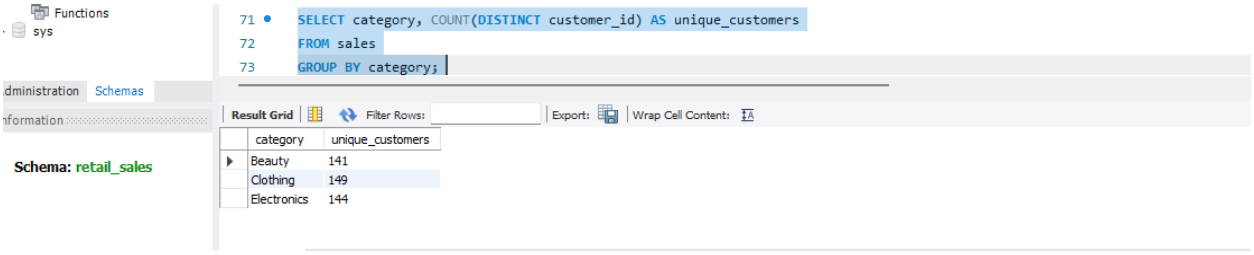
	A	B	C	D	E
1	customer_id	total_spent			
2	3	38440			
3	1	30750			
4	5	30405			
5	2	25295			
6	4	23580			
7					

ix. Unique customers per category

SELECT category, COUNT(DISTINCT customer_id) AS unique_customers

FROM Sales

GROUP BY category;

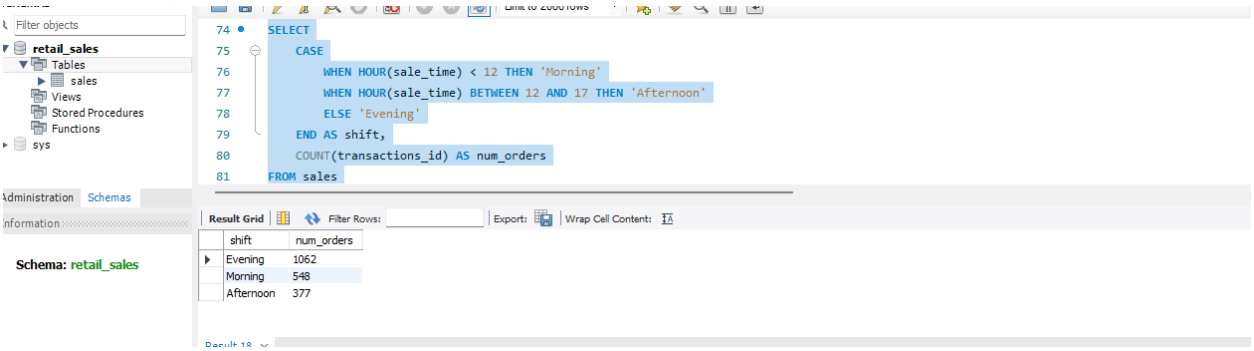


SCREENSHOT OF RESULT ON EXCEL

	A	B	C
1	category	unique_customers	
2	Beauty	141	
3	Clothing	149	
4	Electronic	144	
5			
6			
7			

x. Create shift buckets and count orders

```
SELECT
    CASE
        WHEN HOUR(sale_time) < 12 THEN 'Morning'
        WHEN HOUR(sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'
        ELSE 'Evening'
    END AS shift,
    COUNT(transactions_id) AS num_orders
FROM sales
GROUP BY shift;
```



Key Findings

- Clothing and Electronics were the highest revenue-generating categories.
- High-value transactions (over 1000) were mostly in Electronics.
- November and December showed peak sales (holiday shopping season).
- A few customers contributed disproportionately high sales (top 5 customers).

- Morning and Evening shifts had the most transactions.

Recommendations

1. Stock & Marketing: Focus more inventory on Clothing and Electronics.
2. Customer Segmentation: Target high-value customers with loyalty programs.
3. Seasonality: Prepare for November–December peaks with promotions.
4. Shift Optimization: Increase staff in Morning and Afternoon when sales are highest.
5. Digital Strategy: Encourage digital payments as they are linked to higher-value sales.

Conclusion

The SQL project involved importing retail data, cleaning it, and analyzing it through queries. Business insights revealed top-selling categories, high-value customers, and seasonal trends. This provides actionable recommendations for inventory planning, marketing, and customer management.

The project demonstrates fundamental SQL techniques used by data analysts to explore, clean, and draw insights from real-world retail sales data.

Python Sales Data Analysis Project Report

Introduction

This project demonstrates Python data analysis skills using Pandas and Matplotlib. The dataset represents retail sales transactions. The goal was to clean, explore, and analyze the data to answer specific business questions and extract insights.

Data Importation

The dataset was converted to CSV for easier handling. Imported using Pandas read_csv. Data was previewed with df.head(), number of rows and columns checked and column names checked.

```
[3]: import pandas as pd
[4]: import numpy as np
[5]: import matplotlib.pyplot as plt
[8]: df = pd.read_csv("Python_SalesData.csv")
[9]: print(df.shape)
     (254, 9)
[11]: print (df.columns.tolist() )
     ['Order ID', 'Date', 'Product', 'Price', 'Quantity', 'Purchase Type', 'Payment Method', 'Manager', 'City']
[12]: df = df.loc[:, ~df.columns.astype(str).str.contains("^Unnamed")]
```

Data Cleaning & Preparation

Missing values checked and dropped.

- Data types standardized (dates to datetime, numeric columns to float/int).

- Revenue column created as Quantity * Price.

```
[12]: df = df.loc[:, ~df.columns.astype(str).str.contains("^Unnamed")]
[13]: df.columns = df.columns.str.strip()
      for col in df.select_dtypes(include=['object']).columns:
          df[col] = df[col].astype(str).str.strip()
[14]: df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
      df['Price'] = pd.to_numeric(df['Price'], errors='coerce')
      df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
[15]: df['Revenue'] = df['Price'] * df['Quantity']
```

Exploratory Data Analysis (EDA)

- Descriptive statistics with df.describe().

- Grouping and aggregations by product, city, manager.

- Visualizations using Matplotlib.

Business Questions and Analysis

Q1. Most Preferred Payment Method.

ANSWER: Credit Card=120

```
[17]: df['Payment Method'].value_counts()
```

```
[17]: Payment Method
      Credit Card    120
      Cash            76
      Gift Card       58
      Name: count, dtype: int64
```

Q2. Most Selling Product by quantity and revenue.

ANSWER: Most sold product by Quantity is Beverages=34983.14

```
[43]: product_qty = df.groupby('Product')['Quantity'].sum().reset_index().sort_values(by='Quantity', ascending=False)

[44]: print("Products by Quantity Sold:\n", product_qty)

      Products by Quantity Sold:
                 Product  Quantity
      0          Beverages  34983.14
      3              Fries  32034.34
      1            Burgers  29022.31
      2  Chicken Sandwiches  11135.92
      4       Sides & Other   9819.60

[45]: top_qty = product_qty.iloc[0]

[46]: print(f"\nMost sold product by quantity: {top_qty['Product']} → {top_qty['Quantity']}")

      Most sold product by quantity: Beverages → 34983.14
```

Most sold product by revenue is Burgers=376999.8069

```
[47]: product_revenue = df.groupby('Product')['Revenue'].sum().reset_index().sort_values(by='Revenue',ascending=False)

[48]: print("\nProducts by Revenue:\n", product_revenue)

      Products by Revenue:
                   Product      Revenue
      1            Burgers  376999.8069
      3              Fries  125674.2903
      2  Chicken Sandwiches  114641.6950
      0           Beverages  103200.2630
      4       Sides & Other   48999.8040

[49]: top_rev = product_revenue.iloc[0]

[50]: print(f"Most sold product by revenue: {top_rev['Product']} → {top_rev['Revenue']}")

      Most sold product by revenue: Burgers → 376999.8069
```

Q3. City/Manager with Max Revenue.

ANSWER=City with maximum revenue: Lisbon= 241714.11

Manager with maximum revenue: Joao Silva= 225074.76

```
[35]: top_city = city_revenue.iloc[0]

[36]: print(f"\nCity with maximum revenue: {top_city['City']} → {top_city['Revenue']}")

City with maximum revenue: Lisbon → 241714.11570000002

[37]: top_manager = manager_revenue.iloc[0]

[38]: print(f"Manager with maximum revenue: {top_manager['Manager']} → {top_manager['Revenue']}")

Manager with maximum revenue: Joao Silva → 225074.7665
```

Q4. Average Revenue.

ANSWER=3,029.58

```
[20]: df['Revenue'].mean()

[20]: np.float64(3029.589996850394)
```

Q5. Avg Revenue in Nov–Dec higher (holiday effect).

ANSWER=3,029.58

```
[21]: df[df['Month'].isin([11, 12])]['Revenue'].mean()

[21]: np.float64(3029.589996850394)
```

Q6. Std Dev of Revenue/Quantity indicates variability.

ANSWER=Revenue: 2,420.12 | Quantity: 214.89

```
[22]: df['Revenue'].std(), df['Quantity'].std()

[22]: (2420.11837804107, 214.88869921528863)
```

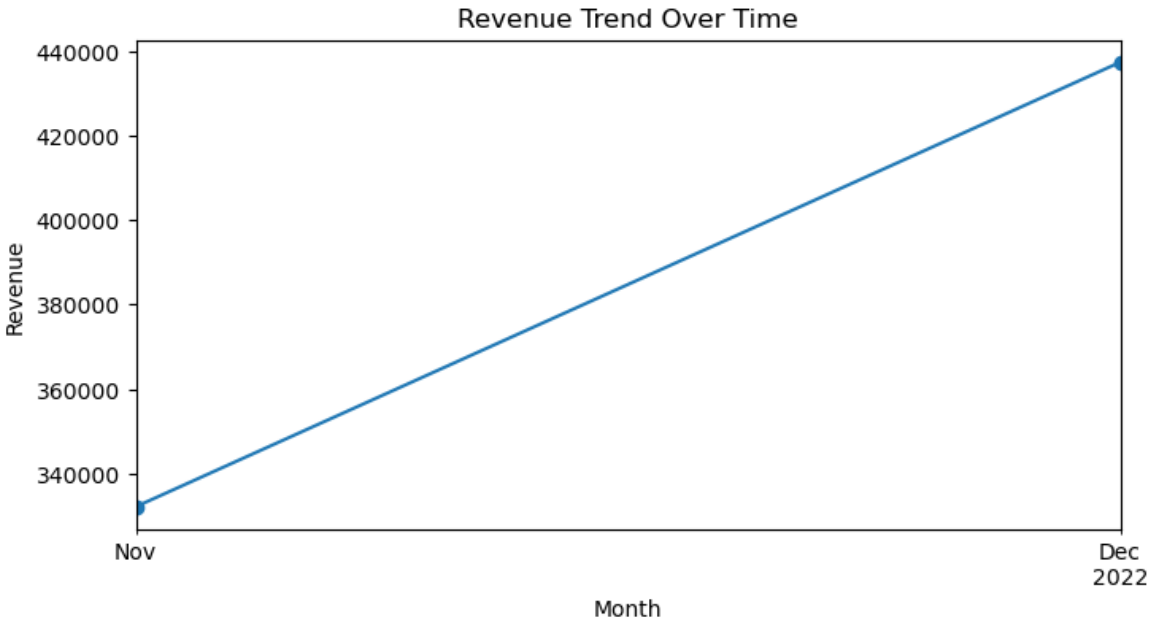
Q7. Variance confirms spread of sales.

Revenue: 5,857,001.11 | Quantity: 46,177.52

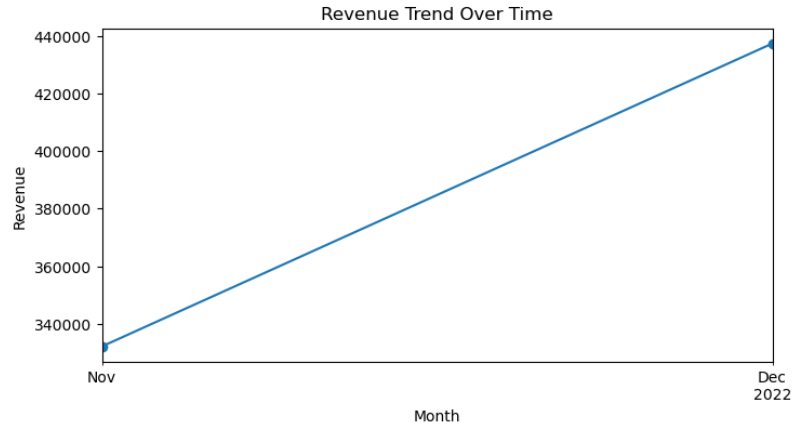
```
[23]: df['Revenue'].var(), df['Quantity'].var()

[23]: (5856972.963732139, 46177.15305043879)
```

Q8. Revenue Trend: steady increase with Q4 peak.




```
[24]: trend = df.groupby(df['Date'].dt.to_period('M'))['Revenue'].sum()
trend.plot(kind='line', marker='o', figsize=(8,4), title="Revenue Trend Over Time")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```



Q9. Avg Quantity & Revenue.

ANSWER

PRODUCT	QUANTITY	REVENUE
Burgers	558.12	7250.0
Fries	628.13	2464.21
Chicken Sandwiches	214.15	2204.6
Beverages	699.66	2064.0
Sides & Other	200.4	1000.0

```
[25]: df.groupby('Product')[['Quantity', 'Revenue']].mean().sort_values('Revenue', ascending=False).head()
```

[25]:

Product	Quantity	Revenue
Burgers	558.121346	7249.996287
Fries	628.124314	2464.201771
Chicken Sandwiches	214.152308	2204.647981
Beverages	699.662800	2064.005260
Sides & Other	200.400000	999.996000

Q10. Total Orders counted.

ANSWER=254

```
[26]: len(df)
```

[26]: 254

Key Findings

- Credit Card is the dominant payment method.
- Beverages = top by volume, Burgers = top by revenue.
- Lisbon generates highest sales.
- November & December peak months.
- Seasonal upward trend in revenue

Recommendations

1. Encourage credit card/digital transactions.

2. Keep Beverages as mass-market product, Burgers as premium.

3. Replicate Lisbon strategies in other cities.

4. Focus ads and inventory in Q4.

5. Train other managers using top performer methods.

Conclusion

The project covered importing, cleaning, exploring, and analyzing sales data.

Using Python libraries, actionable insights were derived to guide data-driven business decisions.

OVERALL PROJECT CONCLUSION

The project confirms that combining multiple analytics tools provides a comprehensive view of business performance.